

Appendix for DrIM: Context-Driven Nearest Neighbor Imputation using Language Representation

A Proof

A.1 Proof of Theorem 1

Proof. [26, 23] showed that the minimizer of $J_{CL}(\theta, \eta)$ is given by

$$r(\mathbf{t}, \mathbf{u}; \theta, \eta) = \log \frac{p(\mathbf{t}, \mathbf{u})}{p_{\mathbf{t}}(\mathbf{t})p_{\mathbf{u}}(\mathbf{u})},$$

for all $\mathbf{t}, \mathbf{u} \in \mathcal{T}$.

Then, for the minimizer of $J_{CL}(\theta, \eta)$, we have

$$\begin{aligned} \left| r(\mathbf{t}, \mathbf{u}; \theta, \eta) - r(\mathbf{t}', \mathbf{u}; \theta, \eta) \right| &= \left| \log \frac{p(\mathbf{t}, \mathbf{u})}{p_{\mathbf{t}}(\mathbf{t})p_{\mathbf{u}}(\mathbf{u})} - \log \frac{p(\mathbf{t}', \mathbf{u})}{p_{\mathbf{t}}(\mathbf{t}')p_{\mathbf{u}}(\mathbf{u})} \right| \\ &= \left| \log \left(\frac{p(\mathbf{t}, \mathbf{u})}{p(\mathbf{t}', \mathbf{u})} \frac{p_{\mathbf{t}}(\mathbf{t}')}{p_{\mathbf{t}}(\mathbf{t})} \right) \right|, \end{aligned}$$

and

$$\begin{aligned} \left| r(\mathbf{t}, \mathbf{u}; \theta, \eta) - r(\mathbf{t}', \mathbf{u}; \theta, \eta) \right| &= \left| g(\mathbf{t}, \mathbf{e}; \theta)^\top g(\mathbf{u}, \mathbf{e}; \theta) + a(g(\mathbf{t}, \mathbf{e}; \theta); \eta_1) + b(g(\mathbf{u}, \mathbf{e}; \theta); \eta_2) \right. \\ &\quad \left. - \left(g(\mathbf{t}', \mathbf{e}; \theta)^\top g(\mathbf{u}, \mathbf{e}; \theta) + a(g(\mathbf{t}', \mathbf{e}; \theta); \eta_1) + b(g(\mathbf{u}, \mathbf{e}; \theta); \eta_2) \right) \right| \\ &= \left| (g(\mathbf{t}, \mathbf{e}; \theta) - g(\mathbf{t}', \mathbf{e}; \theta))^\top g(\mathbf{u}, \mathbf{e}; \theta) + a(g(\mathbf{t}, \mathbf{e}; \theta); \eta_1) - a(g(\mathbf{t}', \mathbf{e}; \theta); \eta_1) \right| \\ &\leq \left| (g(\mathbf{t}, \mathbf{e}; \theta) - g(\mathbf{t}', \mathbf{e}; \theta))^\top g(\mathbf{u}, \mathbf{e}; \theta) \right| + \left| a(g(\mathbf{t}, \mathbf{e}; \theta); \eta_1) - a(g(\mathbf{t}', \mathbf{e}; \theta); \eta_1) \right| \\ &\leq \|g(\mathbf{t}, \mathbf{e}; \theta) - g(\mathbf{t}', \mathbf{e}; \theta)\| \cdot \|g(\mathbf{u}, \mathbf{e}; \theta)\| + L\|g(\mathbf{t}, \mathbf{e}; \theta) - g(\mathbf{t}', \mathbf{e}; \theta)\| \\ &= (\|g(\mathbf{u}, \mathbf{e}; \theta)\| + L) \cdot \|g(\mathbf{t}, \mathbf{e}; \theta) - g(\mathbf{t}', \mathbf{e}; \theta)\|. \end{aligned}$$

The above equations are derived by Cauchy-Schwarz inequality and the Lipschitz continuity.

Since the above inequalities hold for every $\mathbf{u} \in \mathcal{T}$, when the auxiliary sequence $\mathbf{u}$ is given by $\mathbf{e}$, we have

$$\therefore \|g(\mathbf{t}, \mathbf{e}; \theta) - g(\mathbf{t}', \mathbf{e}; \theta)\| \geq \frac{1}{\|g(\mathbf{e}, \mathbf{e}; \theta)\| + L} \cdot \left| \log \frac{p(\mathbf{t}, \mathbf{e})/p(\mathbf{t}', \mathbf{e})}{p_{\mathbf{t}}(\mathbf{t})/p_{\mathbf{t}}(\mathbf{t}')} \right|$$

The proof is complete. $\square$

A.2 Proof of Theorem 2

Proof. [22, 1] showed that

$$\begin{aligned} I(\mathbf{t}; \mathbf{u}) &\geq J_{CL}(\theta) \\ &= \mathbb{E}_{p(\mathbf{t}, \mathbf{u})} [\cos(g(\mathbf{t}, \mathbf{e}; \theta), g(\mathbf{u}, \mathbf{e}; \theta))] - \log \mathbb{E}_{p_{\mathbf{t}}(\mathbf{t})p_{\mathbf{u}}(\mathbf{u}')} [\exp(\cos(g(\mathbf{t}, \mathbf{e}; \theta), g(\mathbf{u}', \mathbf{e}; \theta)))], \end{aligned}$$

where $I(\mathbf{t}; \mathbf{u})$ is the mutual information between $\mathbf{t}$ and $\mathbf{u}$.

Therefore, if we obtain θ such that the maximizer of $J_{CL}(\theta)$, $g(\mathbf{t}, \mathbf{e}; \theta)$ and $g(\mathbf{u}, \mathbf{e}; \theta)$ provide the representation vectors that maximize the lower bound of the mutual information between $\mathbf{t}$ and $\mathbf{u}$.

The proof is complete. $\square$

B Benchmark Datasets

We provide information about the real-world tabular datasets from the UCI repository¹, Kaggle² in our experiment.

B.1 Download links

- abalone [19]: <https://archive.ics.uci.edu/dataset/1/abalone>
- anuran [4]: <https://archive.ics.uci.edu/dataset/406/anuran+calls+mfccs>
- banknote [14]: <https://archive.ics.uci.edu/dataset/267/banknote+authentication>
- breast [31]:
<https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic>
- concrete: [32]: <https://archive.ics.uci.edu/dataset/165/concrete+compressive+strength>
- kings (CC0: Public Domain):
<https://www.kaggle.com/datasets/harlfoxem/housesalesprediction>
- letter [24]: <https://archive.ics.uci.edu/dataset/59/letter+recognition>
- loan (CC0: Public Domain):
<https://www.kaggle.com/datasets/teertha/personal-loan-modeling>
- redwine [5]: <https://archive.ics.uci.edu/dataset/186/wine+quality>
- whitewine [5]: <https://archive.ics.uci.edu/dataset/186/wine+quality>

B.2 Descriptions

Table 1: **Description of datasets.** #C represents the number of continuous variables. #D denotes the number of categorical (discrete) variables. The ‘Target’ refers to the variable used as the response variable in a classification task to evaluate imputation utility.

Dataset	Split	#C	#D	Target	Subject area
abalone	3.3K/0.8K	7	2	Rings	Biology
anuran	5.7K/1.5K	22	3	Species	Computer Science
banknote	1.1K/0.3K	4	1	class	Computer Science
breast	0.5K/0.1K	30	1	Diagnosis	Health and Medicine
concrete	0.8K/0.2K	8	1	Age	Physics and Chemistry
kings	17.3K/4.3K	11	7	grade	Finance
letter	16K/4K	16	1	lettr	Computer Science
loan	4K/1K	5	6	Personal Loan	Finance
redwine	1.3K/0.3K	11	1	quality	Business
whitewine	3.9K/1K	11	1	quality	Business

¹<https://archive.ics.uci.edu/>

²<https://www.kaggle.com/datasets/>

C Reproducibility

We present all the experimental settings for reproducibility in this paper.

C.1 Experimental Settings

- We run experiments using NVIDIA RTX A6000 GPU, and our experimental codes are available with PyTorch and scikit-learn.

Hyperparameters of DrIM: This demonstrates the *generalizability of our proposed model across various tabular datasets*. For all datasets, we applied the following hyperparameters uniformly, without any additional tunings:

- batch size: 16
- k (the number of neighbors): 5
- epochs: 5 (with AdamW optimizer [15])
- learning rate: 5e-5
- #layers (the number of fine-tuning layers): 3

Algorithm 1 Missing data imputation

Input: Incomplete observation $(\mathbf{x}, \mathbf{r})$, Column names $C = \{C_1, C_2, \dots, C_p\}$, Fine-tuned BERT $g(\cdot, \cdot; \theta)$

Output: Complete (imputed) observation: $\hat{\mathbf{x}}$

Phase 1 – Contextual representation

```

1: for  $j = 1, 2, \dots, p$  do
2:   if  $\mathbf{r}_j = 1$  then
3:      $\mathbf{t}_j \leftarrow "C_j \text{ is } \mathbf{x}_j,"$ 
4:   end if
5:   if  $\mathbf{r}_j = 0$  then
6:      $\mathbf{t}_j \leftarrow "C_j \text{ is [MASK],"$  // replacing missing value with [MASK] token
7:   end if
8: end for
9:  $\mathbf{t} \leftarrow "[CLS]" \& \mathbf{t}_1 \& \mathbf{t}_2 \& \dots \& \mathbf{t}_p \& "[SEP]"$ , // Textual encoding
10:  $\mathbf{z} \leftarrow g(\mathbf{t}, \mathbf{e}; \theta) \in \mathbb{R}^D$  // Contextual representation
```

Phase 2 – DRIM

```

11:  $\hat{\mathbf{x}} \leftarrow [0, 0, \dots, 0] \in \mathbb{R}^p$ 
12: for  $j = \{j : \mathbf{r}_j = 0, j = 1, 2, \dots, p\}$  do
13:    $\mathcal{N}_j(\mathbf{x}, k) \subset \{1, 2, \dots, n\}$  // the neighbor index set
14:   if  $j \in I_C$  then
15:      $\hat{\mathbf{x}}_j \leftarrow \frac{1}{k} \sum_{l \in \mathcal{N}_j(\mathbf{x}, k)} \mathbf{x}_j^{(l)}$ 
16:   end if
17:   if  $j \in I_D$  then
18:      $\hat{\mathbf{x}}_j \leftarrow \text{mode}(\mathbf{x}_j^{(l)} : l \in \mathcal{N}_j(\mathbf{x}, k))$ 
19:   end if
20: end for
21:  $\tilde{\mathbf{x}} \leftarrow \mathbf{x} \odot \mathbf{r} + \hat{\mathbf{x}} \odot (1 - \mathbf{r})$ 
```

C.2 Related Works: Missing mechanisms

Descriptions Data missingness is a common challenge in research and practical analysis, categorized into three primary missing mechanisms: 1) Missing Completely at Random (MCAR), 2) Missing at Random (MAR), and 3) Missing Not at Random (MNAR).

Under the **MCAR** mechanism, the reason for missingness is unrelated to any data, whether observed or unobserved. In other words, the likelihood of data being missing is equal across all observations. The primary advantage of MCAR is that it does not introduce bias into the data analysis. However, despite this advantage, data missingness can still reduce the statistical power of the study because of the reduced sample size.

MAR occurs when the probability of missingness is related to the observed data but not the unobserved missing data. Essentially, even though the data is missing, the mechanism assumes that the missingness can be explained by other variables in the dataset. In other words, the missingness can be modeled and imputed using the available data, allowing for more accurate analyses despite the missing values.

Lastly, if the missingness is not specified by either MCAR or MAR, it becomes **MNAR**. MNAR is the most challenging mechanism, as it implies that the missingness is related to the unobserved data itself. In this case, the missing data is systematically different from the observed data, which introduces bias if not properly accounted for. For example, patients with severe symptoms may be less likely to report their health status, resulting in missing data. MNAR requires sophisticated statistical methods to handle, as ignoring or mishandling it can lead to biased and unreliable results.

Implementations Following [18, 34], we generate the missing value mask for each dataset with three mechanisms in four settings, MCAR, MAR, MNARL, and MNARQ.

In the **MCAR** setting, each value is masked according to the realization of a Bernoulli random variable with a fixed parameter.

For the **MAR** setting, a fixed subset of variables that cannot have missing values is sampled in each experiment. The remaining variables are assigned missing values based on a logistic model with random weights, where the non-missing variables serve as inputs. A bias term is optimized via line search to achieve the desired proportion of missing values.

Finally, two different mechanisms are implemented in the MNAR setting. The first, **MNARL**, is identical to the previously described MAR mechanism, but the inputs of the logistic model are then masked by an MCAR mechanism. Hence, the logistic model’s outcome depends on missing values. The second mechanism, **MNARQ**, samples a subset of variables whose values in the lower and upper p th percentiles are masked according to a Bernoulli random variable, and the values in-between are left not missing.

C.3 Details of Implementing Baseline Models

To compare our proposed method with baseline models, we performed experiments across four missing data mechanisms and five missingness rates. Our reproduced codes for the baseline models are provided in the supplementary material. Below are the detailed implementations of the baseline methods:

- Mean [12]: We employ the SimpleImputer package ³, with the `strategy` parameter set to ‘mean’.
- kNNI [28]: We employ the KNNImputer package ⁴ from `scikit-learn` for missing data imputation.
- EM [20]: We employ the EM module from `hyperimpute.Plugins` ⁵.
- SoftImpute [17]: We employ the SoftImpute module in `hyperimpute.Plugins` ⁶.
- MICE [29]: We employ the IterativeImputer package ⁷ from `scikit-learn`.

³<https://scikit-learn.org/1.5/modules/generated/sklearn.impute.SimpleImputer.html>

⁴<https://scikit-learn.org/stable/modules/generated/sklearn.impute.KNNImputer.html>

⁵https://github.com/vanderschaarlab/hyperimpute/blob/main/src/hyperimpute/plugins/imputers/plugin_EM.py

⁶https://github.com/vanderschaarlab/hyperimpute/blob/main/src/hyperimpute/plugins/imputers/plugin_softimpute.py

⁷<https://scikit-learn.org/stable/modules/generated/sklearn.impute.IterativeImputer.html>

- missForest [25]: We employ the missForest module in `hyperimpute.Plugins`⁸.
- Sinkhorn [18]: We utilize the Sinkhorn module in `hyperimpute.Plugins`⁹.
- GAIN [33]: We follow the implementations provided in the official repository¹⁰. As the paper does not explicitly discuss the separate handling of categorical and continuous variables, the code treats them simultaneously. Consequently, a rounding process is employed to handle categorical variables afterward.
- VAEAC [10]: We follow the implementations provided in the official repository¹¹. The authors provided hyperparameters that adequately address both continuous and categorical variables, so we used these without further modification during model fitting.
- MIWAE [16]: The implemented MIWAE code in the official repository¹² was designed for continuous variables only. To accommodate heterogeneous tabular datasets, we treat the conditional distribution of categorical columns as categorical distributions and employ cross-entropy loss for reconstruction.
- not-MIWAE [9]: The implemented not-MIWAE code in the official repository¹³ also focused on continuous variables exclusively. To handle categorical variables, we make the same modifications as in MIWAE.
- MIRACLE [11]: We utilize the MIRACLE module in `hyperimpute.Plugins`¹⁴.
- ReMasker [7]: We follow the implementations provided in official repository¹⁵.

C.4 Evaluation settings

C.4.1 Imputation fidelity

ASMAPE The average symmetric mean absolute percentage error (ASMAPE), which combines the symmetric mean absolute percentage error (SMAPE) for continuous variables and the accuracy error (ERR) for categorical variables.

$$\text{ASMAPE}(\mathbf{x}, \tilde{\mathbf{x}}) = \underbrace{\frac{1}{\sum_{j \in I_C} (1 - \mathbf{r}_j)} \sum_{j \in I_C} \frac{|\mathbf{x}_j - \tilde{\mathbf{x}}_j|}{|\mathbf{x}_j| + |\tilde{\mathbf{x}}_j|}}_{\text{SMAPE}} + \underbrace{\frac{1}{\sum_{j \in I_D} (1 - \mathbf{r}_j)} \sum_{j \in I_D} \mathbb{I}(\mathbf{x}_j \neq \tilde{\mathbf{x}}_j)}_{\text{ERR}}$$

where $\mathbb{I}(\cdot)$ is the indicator function.

C.4.2 Imputation utility

Regarding **imputation utility**, we conduct classification, model selection, and feature selection tasks for post-imputation evaluation. Please refer to Table 1 for the classification target variable, and Table 2 for detailed machine learning model configuration. The detailed evaluation procedure is as follows:

⁸https://github.com/vanderschaarlab/hyperimpute/blob/main/src/hyperimpute/plugins/imputers/plugin_missforest.py

⁹https://github.com/vanderschaarlab/hyperimpute/blob/main/src/hyperimpute/plugins/imputers/plugin_sinkhorn.py

¹⁰<https://github.com/jsyo0823/GAIN/tree/master>

¹¹<https://github.com/tigvarts/vaeac>

¹²<https://github.com/pamattei/miwae>

¹³<https://github.com/nbip/notMIWAE>

¹⁴https://github.com/vanderschaarlab/hyperimpute/blob/main/src/hyperimpute/plugins/imputers/plugin_miracle.py

¹⁵<https://github.com/tydusky/remasker>

Table 2: **Classifier used to evaluate imputed data quality in imputation utility.** The names of all parameters used in the description are consistent with those defined in the corresponding packages.

Task	Model	Description
Classification	Logistic Regression	Package: <code>sklearn.linear_model.LogisticRegression</code> , setting: <code>random_state=0</code> , <code>max_iter=1000</code> , defaulted values
	Gaussian Naive Bayes	Package: <code>sklearn.naive_bayes.GaussianNB</code> , setting: defaulted values
	k-Nearest Neighbors	Package: <code>sklearn.neighbors.KNeighborsClassifier</code> , setting: defaulted values
	Decision Tree	Package: <code>sklearn.tree.DecisionTreeClassifier</code> , setting: <code>random_state=0</code> , defaulted values
	Random Forest	Package: <code>sklearn.ensemble.RandomForestClassifier</code> , setting: <code>random_state=0</code> , defaulted values

Classification performance (F_1)

1. Train an imputer using a given incomplete train dataset.
2. Impute the incomplete train dataset using the trained imputer.
3. Train five machine learning models in Table 2 using the imputed dataset.
4. Assess classification prediction performance by averaging the F_1 scores from the complete test dataset from five different classifiers.

Model selection performance (Model)

1. Train an imputer using a given incomplete train dataset.
2. Impute the incomplete train dataset using the trained imputer.
3. Train five machine learning models in Table 2 using both the original complete train dataset (before generating missing data) and the imputed dataset.
4. Evaluate the classification performance (AUROC) of all trained classifiers on the complete test dataset.
5. Assess model selection performance by comparing the AUROC rank orderings of classifiers trained on the original complete train dataset and those trained on the imputed dataset using Spearman’s Rank Correlation.

Feature selection performance (Feature)

1. Train an imputer using a given incomplete train dataset.
2. Impute the incomplete train dataset using the trained imputer.
3. Train a Random Forest classifier using both the original complete train dataset and the imputed dataset.
4. Determine the rank-ordering of important features for both classifiers.
5. Assess feature selection performance by comparing the feature importance rank orderings of classifiers trained on the original complete train dataset and those trained on the imputed dataset using Spearman’s Rank Correlation.

C.5 Language Models

We provide detailed information on the language model configurations and the application of contrastive learning to autoregressive models.

C.5.1 Configurations

We provide the configurations of the language model used in RQ4.

Table 3: **The configurations of the language models** used to generate contextual representations. ‘#Params’ represents the number of parameters, as defined in the specifications of each language model.

Type	model	#Params
Autoencoding	BERT _{BASE} [6]	0.11B
	RoBERTa [13]	0.12B
	BERT _{LARGE} [6]	0.34B
Autoregressive	GPT-2 [21]	1.56B
	GPT-Neo [2]	1.32B
	LLaMA [27]	6.61B

C.5.2 Modifications for the autoregressive model

We make slight modifications for contrastive learning using the autoregressive model as follows:

- **Missing value representation:** Since autoregressive models do not include the [MASK] token, we use the [UNK] token to represent missing values in these models.
- **Representation vector of autoregressive model** (Definition 1): We aggregate contextual information from all tokens in the sequence by applying average pooling to the last hidden layer, which is then used as the representation vector, since there is no special token like the [CLS] token for aggregation.

D Additional experiments

D.1 Overview

In this section, we present additional experimental results that could not be included in the main text due to space limitations. The experiments fall into three categories. First, we conduct an ablation study to evaluate the importance of each component of the proposed model. Second, we examine the scalability and efficiency of kNN-based methods and large language models, which typically incur substantial computational costs. Third, we compare different strategies for converting tabular data into natural language. Below, we summarize the setup and key findings of each experiment.

1. Scalability and efficiency
2. Various distance functions
3. How to convert tabular data into natural language

D.2 Scalability and efficiency

We conducted additional experiments on two datasets with more than 100 variables:

- **nomao** [3]: 34465 samples, 89 continuous features, 30 categorical features, classification target: Class.
Download link: https://www.openml.org/search?type=data&sort=runs&status=active&qualities.NumberOfFeatures=between_100_1000&order=desc&id=1486
- **yeast** [8]: 2417 samples, 103 continuous features, 14 categorical features, classification target: class14.
Download link: https://www.openml.org/search?type=data&sort=runs&status=active&qualities.NumberOfFeatures=between_100_1000&id=316

Table 4: Imputation performance at a **0.3** missingness rate under **MAR**. The means and their standard errors across 10 repeated experiments are reported. $\uparrow$ ($\downarrow$) denotes that higher (lower) values are better. The best value is **bolded**. † Imputation could not be performed due to model failure.

model	yeast					nomao				
	ASMAPE $\downarrow$	KL $\downarrow$	F_1 $\uparrow$	Model $\uparrow$	Feature $\uparrow$	ASMAPE $\downarrow$	KL $\downarrow$	F_1 $\uparrow$	Model $\uparrow$	Feature $\uparrow$
Mean	1.240 $\pm$.014	.060 $\pm$.001	.905 $\pm$.006	.891 $\pm$.037	.646 $\pm$.024	.715 $\pm$.014	.227 $\pm$.016	.812 $\pm$.013	.610 $\pm$.110	.809 $\pm$.012
kNNI	.866 $\pm$.014	.008 $\pm$.000	.914 $\pm$.005	.911 $\pm$.032	.685 $\pm$.023	.647 $\pm$.013	.234 $\pm$.004	.814 $\pm$.013	.740 $\pm$.052	.834 $\pm$.013
EM	1.193 $\pm$.050	.055 $\pm$.005	.828 $\pm$.040	.879 $\pm$.035	.644 $\pm$.031	.823 $\pm$.082	.253 $\pm$.024	.729 $\pm$.028	.656 $\pm$.121	.810 $\pm$.012
SoftImpute	1.044 $\pm$.018	.031 $\pm$.001	.928 $\pm$.005	.889 $\pm$.037	.673 $\pm$.024	.712 $\pm$.014	.561 $\pm$.016	.820 $\pm$.013	.720 $\pm$.095	.722 $\pm$.014
MICE	.880 $\pm$.014	.006 $\pm$.000	.909 $\pm$.006	.882 $\pm$.037	.692 $\pm$.019	.674 $\pm$.014	.469 $\pm$.014	.829 $\pm$.013	.730 $\pm$.100	.693 $\pm$.021
missForest	.910 $\pm$.014	.024 $\pm$.000	.925 $\pm$.004	.911 $\pm$.032	.665 $\pm$.026	.670 $\pm$.014	.262 $\pm$.012	.817 $\pm$.013	.770 $\pm$.042	.706 $\pm$.014
Sinkhorn	.884 $\pm$.013	.016 $\pm$.000	.922 $\pm$.005	.912 $\pm$.029	.674 $\pm$.027	.692 $\pm$.014	.482 $\pm$.012	.822 $\pm$.012	.697 $\pm$.092	.679 $\pm$.023
GAIN	.885 $\pm$.014	.013 $\pm$.000	.918 $\pm$.004	.920 $\pm$.030	.668 $\pm$.022	.681 $\pm$.014	.484 $\pm$.013	.821 $\pm$.014	.687 $\pm$.139	.666 $\pm$.021
VAEAC	.882 $\pm$.007	.004 $\pm$.000	.929 $\pm$.007	.897 $\pm$.038	.671 $\pm$.027	.168 $\pm$.002	.342 $\pm$.015	.822 $\pm$.001	.827 $\pm$.039	.878 $\pm$.006
MIWAE	1.002 $\pm$.013	.005 $\pm$.000	.912 $\pm$.005	.914 $\pm$.028	.667 $\pm$.024	-	-	OOM	-	-
not-MIWAE	.852 $\pm$.014	.009 $\pm$.000	.918 $\pm$.004	.897 $\pm$.038	.692 $\pm$.024	.670 $\pm$.002	.734 $\pm$.010	.846 $\pm$.002	.720 $\pm$.073	.772 $\pm$.010
MIRACLE	1.257 $\pm$.018	.009 $\pm$.001	.790 $\pm$.001	.789 $\pm$.141	.480 $\pm$.038	-	-	N/A †	-	-
ReMasker	.963 $\pm$.014	.023 $\pm$.001	.926 $\pm$.004	.920 $\pm$.030	.664 $\pm$.023	.656 $\pm$.014	.423 $\pm$.015	.824 $\pm$.013	.690 $\pm$.103	.683 $\pm$.019
DrIM	.826 $\pm$.006	.004 $\pm$.000	.933 $\pm$.004	.903 $\pm$.043	.693 $\pm$.028	.247 $\pm$.010	.182 $\pm$.009	.903 $\pm$.003	.840 $\pm$.040	.940 $\pm$.004

Table 5: Computational cost in terms of imputation (training + inference) time and memory consumption. The means across 10 repeated experiments are reported.

dataset	Mean	kNNI	EM	SoftImpute	MICE	missForest	Sinkhorn	GAIN	VAEAC	MIWAE	not-MIWAE	MIRACLE	ReMasker	DrIM
yeast														
Time (m)	< 1	< 1	< 1	2	5	6	11	< 1	65	21	66	8	3	12
memory (MB)	-	-	-	387	2160	422	478	2644	3545	2846	2688	2081	2755	3155
nomao														
Time (m)	< 1	8	2	20	160	37	12	< 1	210	-	1180	-	36	168
memory (MB)	-	2648	6864	784	5004	917	850	2850	3593	-	3657	-	3859	12329

We provide below our findings regarding both computational cost and missing data imputation performance for all baseline models.

We fine-tune only three BERT layers instead of all, which yields faster training and inference than some baselines (VAEAC, MIWAE, not-MIWAE). Although our language model-based approach uses more memory, it delivers strong imputation performance, with superior results in imputation utility (F_1 , Model, Feature) and imputation fidelity (KL) metrics.

It is worth noting that AI models, in general, tend to consume more resources compared to traditional machine learning models. Based on the results we have obtained, we consider the additional computational cost to be a reasonable trade-off for improving missing data imputation performance.

D.3 Various distance functions

We investigated imputation quality using Manhattan distance, Mahalanobis distance, Chebyshev distance, and correlation measures. The following results illustrate the outcomes of these evaluations.

Table 6: Imputation performance based on different distance functions at a 0.3 missingness rate under MAR. The means and their standard errors across 10 datasets and 10 repeated experiments are reported. $\uparrow$ ($\downarrow$) denotes that higher (lower) values are better.

Distance	ASMAPE $\downarrow$	KL $\downarrow$	F_1 $\uparrow$	Model $\uparrow$	Feature $\uparrow$
Manhattan	.790 $\pm$.028	.019 $\pm$.002	.659 $\pm$.024	.656 $\pm$.031	.896 $\pm$.009
Mahalanobis	.850 $\pm$.031	.044 $\pm$.004	.647 $\pm$.024	.621 $\pm$.036	.876 $\pm$.012
Chebyshev	.797 $\pm$.027	.019 $\pm$.002	.663 $\pm$.024	.639 $\pm$.033	.900 $\pm$.008
Correlation	.788 $\pm$.027	.019 $\pm$.002	.662 $\pm$.024	.637 $\pm$.033	.893 $\pm$.010
Cosine	.774 $\pm$.022	.018 $\pm$.001	.663 $\pm$.024	.638 $\pm$.033	.902 $\pm$.009

Even when the data is mixed-type, the layer normalization functionality of BERT ensures that the [CLS] token’s representation vector resides on a sphere. In fact, the mean and standard deviation (SD) of the BERT representation vectors for each dataset we used are as follows:

As a result, the Mahalanobis distance fails to capture this representation effectively. In fact, when the data points are positioned on the sphere, distances like Manhattan, Chebyshev, Correlation, and

Table 7: Mean and standard deviation (SD) of BERT [CLS] representation vectors across datasets.

Dataset	abalone	anuran	banknote	breast	concrete	kings	letter	loan	redwine	whitewine
Mean	11.46	11.31	12.54	10.07	11.32	13.53	10.69	9.34	12.45	8.53
SD	1.44	0.68	2.11	0.47	1.65	0.02	0.21	0.11	0.17	0.14

Cosine similarity tend to form very similar neighbor structures. For example, if the Euclidean distance between two neighbors is $d > 0$, the cosine similarity we selected can be expressed as $1 - \frac{1}{2}d^2$, which essentially allows us to find neighbors in the same way as using Euclidean distance. Moreover, the covariance of the [CLS] token’s representation vector, which is needed for calculating Mahalanobis distance, often turns out to be singular, leading to significant instability in the calculations.

D.4 How to convert tabular data into natural language

We conducted an empirical study to assess the effectiveness of converting tabular data into natural language format. In addition to using textual encoding and direct numerical embedding methods, we incorporated the JSON formatting approach presented in [30]. Our experiments demonstrated the following results.

Table 8: Imputation performance based on different encoding strategies at a 0.3 missingness rate under MAR. The means and their standard errors across 10 datasets and 10 repeated experiments are reported. $\uparrow$ ($\downarrow$) denotes that higher (lower) values are better.

encoding	ASMAPE $\downarrow$	KL $\downarrow$	F_1 $\uparrow$	Model $\uparrow$	Feature $\uparrow$
Direct Numerical embedding	.789 $\pm$.027	.019 $\pm$.001	.663 $\pm$.024	.667 $\pm$.031	.900 $\pm$.008
JSON format encoding	.807 $\pm$.029	.019 $\pm$.001	.659 $\pm$.025	.666 $\pm$.028	.884 $\pm$.008
Textual encoding	.774 $\pm$.022	.018 $\pm$.001	.663 $\pm$.024	.638 $\pm$.033	.902 $\pm$.009

Our experimental results indicate that our textual encoding method is effective across all metrics except for the model selection metric. However, given that its performance is not substantially different from that of direct numerical embedding, we expect there is still room to explore alternative methods for converting tabular data for language model input. We plan to investigate this further in future research.

E Detailed experimental results

E.1 Imputation fidelity

In this section, we present the imputation fidelity performance across 0.2, 0.4, 0.6, and 0.8 missingness rates.

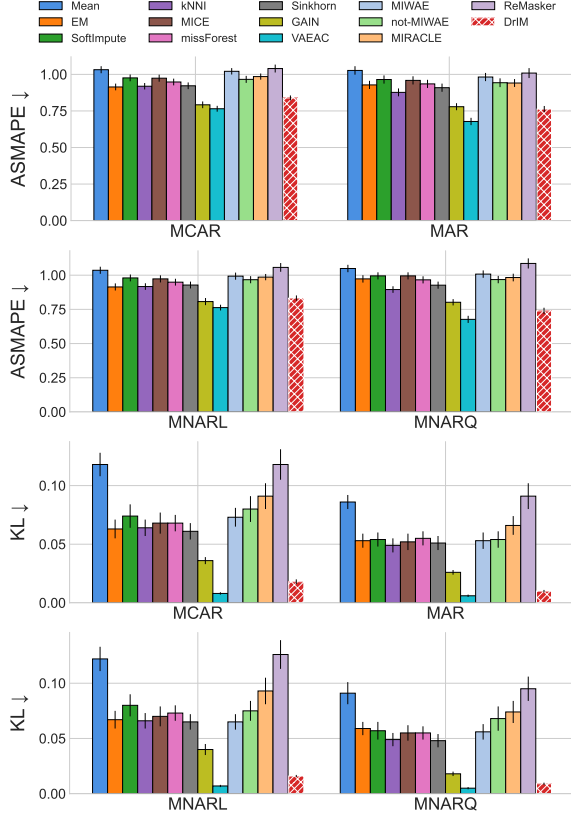


Figure 1: Imputation fidelity at **0.2** missingness rate. The corresponding missing mechanism is indicated below the figure. The means and their standard errors across 10 datasets and 10 repeated experiments are reported. ↓ denotes lower is better.

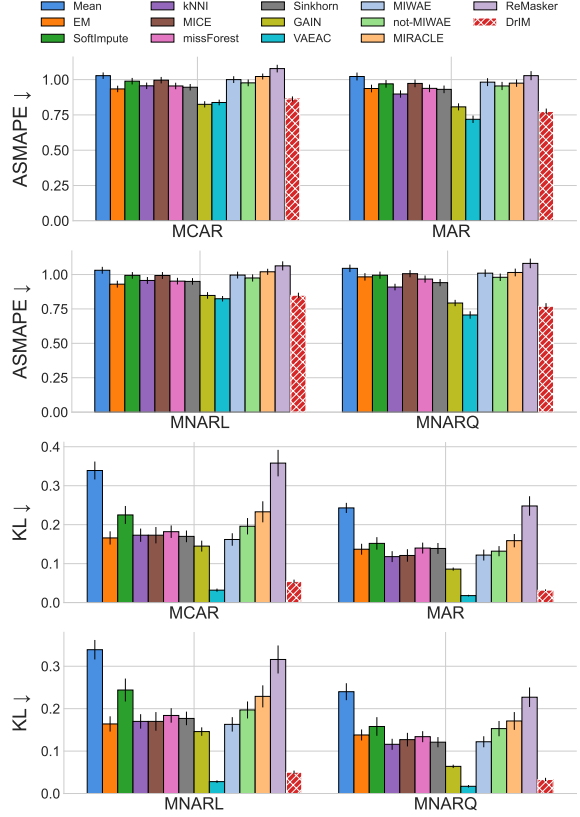


Figure 2: Imputation fidelity at **0.4** missingness rate. The corresponding missing mechanism is indicated below the figure. The means and their standard errors across 10 datasets and 10 repeated experiments are reported. ↓ denotes lower is better.

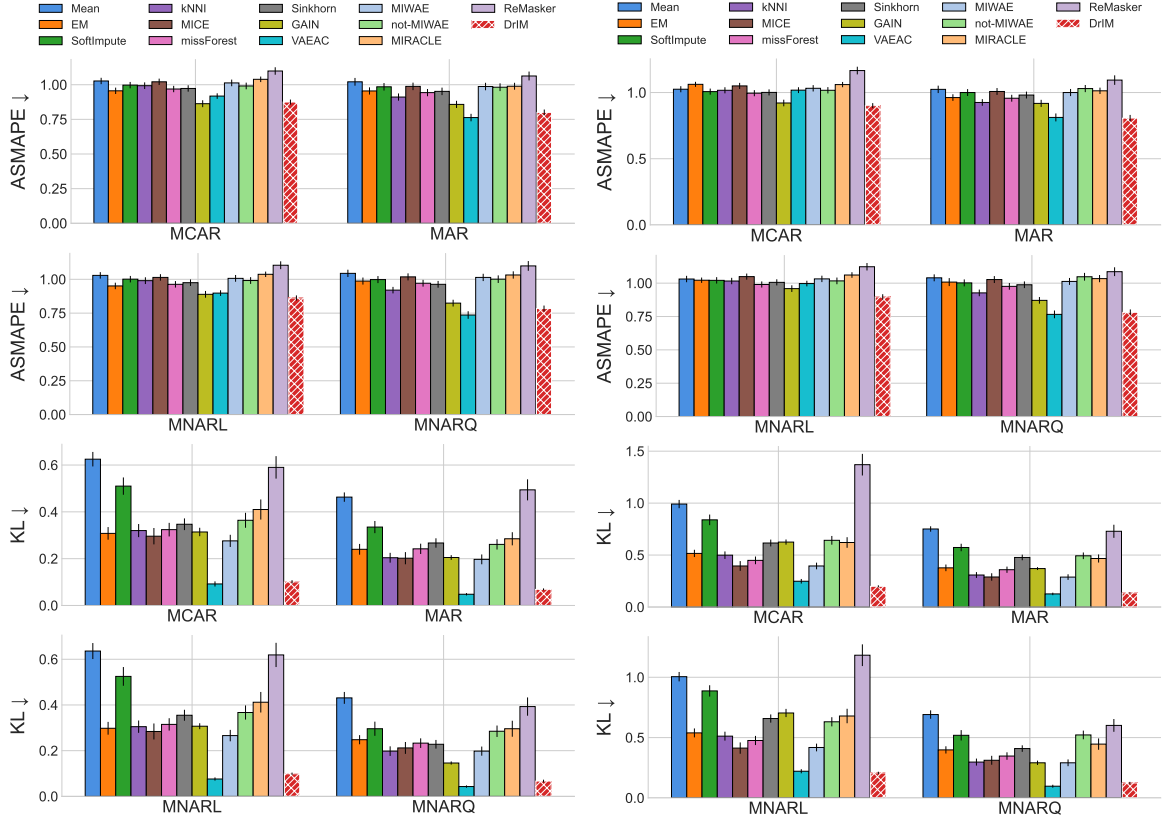


Figure 3: Imputation fidelity at **0.6** missingness rate. The corresponding missing mechanism is indicated below the figure. The means and their standard errors across 10 datasets and 10 repeated experiments are reported. $\downarrow$ denotes lower is better.

Figure 4: Imputation fidelity at **0.8** missingness rate. The corresponding missing mechanism is indicated below the figure. The means and their standard errors across 10 datasets and 10 repeated experiments are reported. $\downarrow$ denotes lower is better.

The detailed experimental results in Appendix E.1 confirm that DRIM achieves high imputation fidelity across a variety of missing mechanisms and rates. In these tables, two main performance indicators are reported:

- **ASMAPE (Average Symmetric Mean Absolute Percentage Error):** DRIM consistently yields lower ASMAPE values than baseline methods, indicating that its imputed values are closer to the ground-truth on a per-entry basis. The reduction in error is not marginal; as missingness increases (e.g., from 0.3 to 0.8), DRIM’s error remains significantly lower compared to traditional methods such as kNNI and SoftImpute. This suggests that the model is effectively capturing and reconstructing the local data structure, even under high missingness conditions.
- **KL-Divergence:** KL divergence values are uniformly lower for DRIM, demonstrating that the statistical distribution of the imputed dataset aligns well with the original data distribution. The preservation of global distribution characteristics is critical for downstream tasks.

These results together imply that DRIM is effective at maintaining both the value-level accuracy and the global structure of the data.

E.2 Imputation utility

In this section, we provide the imputation utility performance across various missingness rates, including 0.2, 0.4, 0.6, and 0.8.

Table 9: Imputation utility at **0.2** missingness rate. The means and their standard errors across 10 datasets and 10 repeated experiments are reported. $\uparrow$ denotes higher is better. The best value is **bolded**, and the second best is underlined.

Model	MCAR			MAR			MNARL			MNARQ		
	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$
Mean	.646 $\pm$.026	.543 $\pm$.037	.936 $\pm$.005	.642 $\pm$.026	<u>.710</u> $\pm$.030	.863 $\pm$.013	.629 $\pm$.026	.623 $\pm$.034	.904 $\pm$.009	.668 $\pm$.026	.651 $\pm$.034	.899 $\pm$.009
EM	.657 $\pm$.025	.523 $\pm$.032	.934 $\pm$.007	.641 $\pm$.026	.707 $\pm$.037	.912 $\pm$.010	.640 $\pm$.026	.619 $\pm$.038	.886 $\pm$.012	.669 $\pm$.025	.623 $\pm$.032	.916 $\pm$.008
SoftImpute	.650 $\pm$.026	.530 $\pm$.035	.939 $\pm$.005	.643 $\pm$.026	.694 $\pm$.032	.905 $\pm$.009	.631 $\pm$.026	.576 $\pm$.036	.903 $\pm$.009	.669 $\pm$.026	.670 $\pm$.031	.932 $\pm$.006
kNNI	.654 $\pm$.025	.572 $\pm$.032	.940 $\pm$.008	.649 $\pm$.025	.705 $\pm$.032	.923 $\pm$.010	.636 $\pm$.026	.605 $\pm$.039	.899 $\pm$.012	.672 $\pm$.026	.646 $\pm$.032	.930 $\pm$.007
MICE	.656 $\pm$.026	.489 $\pm$.035	.946 $\pm$.005	.648 $\pm$.026	.690 $\pm$.034	.912 $\pm$.009	.637 $\pm$.026	.601 $\pm$.036	.904 $\pm$.010	.673 $\pm$.026	.633 $\pm$.033	<u>.933</u> $\pm$.007
missForest	.651 $\pm$.026	.557 $\pm$.033	.925 $\pm$.007	.646 $\pm$.026	.705 $\pm$.032	.903 $\pm$.009	.634 $\pm$.026	.595 $\pm$.039	.894 $\pm$.011	.671 $\pm$.026	.659 $\pm$.033	.920 $\pm$.007
Sinkhorn	.651 $\pm$.026	.563 $\pm$.035	.932 $\pm$.008	.649 $\pm$.025	.696 $\pm$.033	.911 $\pm$.009	.635 $\pm$.026	.617 $\pm$.037	.900 $\pm$.011	.671 $\pm$.026	.649 $\pm$.033	.918 $\pm$.008
GAIN	<u>.669</u> $\pm$.025	<u>.659</u> $\pm$.022	.935 $\pm$.005	<u>.673</u> $\pm$.025	.682 $\pm$.025	<u>.931</u> $\pm$.006	<u>.662</u> $\pm$.024	<u>.630</u> $\pm$.032	.914 $\pm$.008	<u>.685</u> $\pm$.025	<u>.682</u> $\pm$.026	.931 $\pm$.005
VAEAC	.611 $\pm$.025	.634 $\pm$.019	.857 $\pm$.004	.610 $\pm$.017	.853 $\pm$.005	.609 $\pm$.025	.626 $\pm$.019	.851 $\pm$.005	.614 $\pm$.024	.679 $\pm$.014	.847 $\pm$.008	
MIWAE	.632 $\pm$.026	.581 $\pm$.034	.931 $\pm$.007	.641 $\pm$.026	.669 $\pm$.034	.902 $\pm$.009	.629 $\pm$.026	.578 $\pm$.039	.896 $\pm$.011	.667 $\pm$.026	.645 $\pm$.034	.929 $\pm$.006
not-MIWAE	.647 $\pm$.025	.593 $\pm$.032	.939 $\pm$.006	.643 $\pm$.026	.663 $\pm$.035	.904 $\pm$.010	.628 $\pm$.026	.597 $\pm$.037	.898 $\pm$.010	.669 $\pm$.026	.669 $\pm$.032	.932 $\pm$.006
MIRACLE	.648 $\pm$.025	.474 $\pm$.038	.937 $\pm$.006	.646 $\pm$.026	.609 $\pm$.039	.911 $\pm$.009	.630 $\pm$.026	.549 $\pm$.041	.889 $\pm$.012	.670 $\pm$.026	.623 $\pm$.033	.923 $\pm$.008
ReMasker	.648 $\pm$.026	.493 $\pm$.037	.907 $\pm$.010	.643 $\pm$.026	.646 $\pm$.037	.888 $\pm$.014	.630 $\pm$.027	.523 $\pm$.041	.870 $\pm$.013	.666 $\pm$.026	.599 $\pm$.036	.919 $\pm$.008
DrIM	.670 $\pm$.024	.719 $\pm$.025	<u>.945</u> $\pm$.005	.681 $\pm$.025	.754 $\pm$.026	.937 $\pm$.007	.666 $\pm$.024	.679 $\pm$.029	.937 $\pm$.006	.692 $\pm$.025	.785 $\pm$.020	.945 $\pm$.005

Table 10: Imputation utility at **0.4** missingness rate. The means and their standard errors across 10 datasets and 10 repeated experiments are reported. $\uparrow$ denotes higher is better. The best value is **red bolded**, and the second best is underlined.

Model	MCAR			MAR			MNARL			MNARQ		
	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$
Mean	.496 $\pm$.030	.062 $\pm$.046	.672 $\pm$.037	.533 $\pm$.028	.370 $\pm$.047	.607 $\pm$.029	.493 $\pm$.029	.201 $\pm$.050	.662 $\pm$.031	.561 $\pm$.030	.293 $\pm$.057	.632 $\pm$.030
EM	.532 $\pm$.028	.093 $\pm$.052	.819 $\pm$.018	.540 $\pm$.028	.369 $\pm$.050	.780 $\pm$.020	.522 $\pm$.028	.179 $\pm$.050	.733 $\pm$.024	.578 $\pm$.030	.302 $\pm$.056	.767 $\pm$.024
SoftImpute	.491 $\pm$.030	-.080 $\pm$.045	.740 $\pm$.025	.534 $\pm$.029	.285 $\pm$.049	.743 $\pm$.024	.491 $\pm$.029	.118 $\pm$.052	.715 $\pm$.022	.563 $\pm$.030	.240 $\pm$.058	.785 $\pm$.023
kNNI	.529 $\pm$.028	.087 $\pm$.053	.849 $\pm$.016	.547 $\pm$.028	.343 $\pm$.051	.806 $\pm$.018	.521 $\pm$.027	.238 $\pm$.055	.775 $\pm$.021	.576 $\pm$.030	.271 $\pm$.055	.806 $\pm$.021
MICE	.506 $\pm$.030	-.032 $\pm$.049	<u>.851</u> $\pm$.015	.539 $\pm$.029	.319 $\pm$.053	<u>.817</u> $\pm$.017	.501 $\pm$.029	.084 $\pm$.052	.774 $\pm$.020	.569 $\pm$.031	.265 $\pm$.057	<u>.816</u> $\pm$.020
missForest	.509 $\pm$.029	-.001 $\pm$.044	.791 $\pm$.021	.541 $\pm$.028	.329 $\pm$.050	.775 $\pm$.019	.503 $\pm$.029	.182 $\pm$.049	.731 $\pm$.022	.570 $\pm$.030	.266 $\pm$.056	.779 $\pm$.022
Sinkhorn	.511 $\pm$.029	.001 $\pm$.047	.798 $\pm$.021	.540 $\pm$.028	.341 $\pm$.048	.776 $\pm$.021	.505 $\pm$.028	.173 $\pm$.049	.748 $\pm$.022	.572 $\pm$.030	.271 $\pm$.056	.748 $\pm$.027
GAIN	.587 $\pm$.026	.320 $\pm$.044	.725 $\pm$.028	<u>.612</u> $\pm$.026	.475 $\pm$.036	.781 $\pm$.019	.583 $\pm$.026	<u>.250</u> $\pm$.043	.726 $\pm$.022	<u>.638</u> $\pm$.026	.461 $\pm$.040	.794 $\pm$.019
VAEAC	<u>.595</u> $\pm$.025	<u>.488</u> $\pm$.026	.789 $\pm$.013	.598 $\pm$.025	<u>.521</u> $\pm$.025	.790 $\pm$.012	<u>.592</u> $\pm$.025	.495 $\pm$.025	<u>.788</u> $\pm$.014	.603 $\pm$.025	<u>.585</u> $\pm$.023	.782 $\pm$.016
MIWAE	.491 $\pm$.030	-.041 $\pm$.047	.791 $\pm$.021	.530 $\pm$.028	.317 $\pm$.051	.781 $\pm$.019	.487 $\pm$.029	.083 $\pm$.049	.739 $\pm$.021	.560 $\pm$.030	.236 $\pm$.056	.790 $\pm$.021
not-MIWAE	.490 $\pm$.030	-.074 $\pm$.050	.798 $\pm$.020	.529 $\pm$.028	.347 $\pm$.049	.770 $\pm$.020	.486 $\pm$.029	.103 $\pm$.054	.746 $\pm$.019	.559 $\pm$.031	.246 $\pm$.058	.780 $\pm$.022
MIRACLE	.499 $\pm$.029	-.055 $\pm$.049	.812 $\pm$.015	.535 $\pm$.028	.277 $\pm$.058	.769 $\pm$.021	.493 $\pm$.028	.029 $\pm$.051	.737 $\pm$.024	.566 $\pm$.030	.203 $\pm$.054	.797 $\pm$.023
ReMasker	.496 $\pm$.031	.081 $\pm$.050	.750 $\pm$.026	.535 $\pm$.029	.341 $\pm$.048	.718 $\pm$.028	.499 $\pm$.029	.104 $\pm$.054	.703 $\pm$.027	.557 $\pm$.031	.249 $\pm$.050	.759 $\pm$.025
DrIM	.620 $\pm$.024	.493 $\pm$.034	.871 $\pm$.011	.643 $\pm$.024	.544 $\pm$.038	.843 $\pm$.016	.611 $\pm$.024	.495 $\pm$.025	.841 $\pm$.014	.657 $\pm$.025	.617 $\pm$.035	.851 $\pm$.015

Table 11: Imputation utility at **0.6** missingness rate. The means and their standard errors across 10 datasets and 10 repeated experiments are reported. $\uparrow$ denotes higher is better. The best value is **red bolded**, and the second best is underlined.

Model	MCAR			MAR			MNARL			MNARQ		
	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$
Mean	.377 $\pm$.031	-.186 $\pm$.051	.422 $\pm$.047	.433 $\pm$.030	.012 $\pm$.059	.411 $\pm$.037	.380 $\pm$.030	-.138 $\pm$.048	.427 $\pm$.043	.464 $\pm$.034	.034 $\pm$.063	.450 $\pm$.041
EM	.429 $\pm$.030	-.255 $\pm$.047	.669 $\pm$.025	.461 $\pm$.029	.024 $\pm$.054	.626 $\pm$.029	.432 $\pm$.029	-.175 $\pm$.049	.569 $\pm$.033	.494 $\pm$.033	.024 $\pm$.059	.567 $\pm$.034
SoftImpute	.363 $\pm$.030	-.277 $\pm$.053	.445 $\pm$.045	.432 $\pm$.031	-.027 $\pm$.060	.469 $\pm$.043	.374 $\pm$.030	-.226 $\pm$.048	.472 $\pm$.041	.462 $\pm$.034	-.006 $\pm$.064	.567 $\pm$.037
kNNI	.441 $\pm$.029	-.151 $\pm$.058	.749 $\pm$.020	.458 $\pm$.030	.033 $\pm$.057	.677 $\pm$.024	.439 $\pm$.029	-.100 $\pm$.054	.609 $\pm$.027	.486 $\pm$.033	.074 $\pm$.061	.659 $\pm$.031
MICE	.361 $\pm$.031	-.295 $\pm$.052	<u>.732</u> $\pm$.019	.441 $\pm$.031	-.001 $\pm$.055	<u>.684</u> $\pm$.026	.378 $\pm$.031	-.247 $\pm$.052	.624 $\pm$.028	.468 $\pm$.034	.029 $\pm$.061	.673 $\pm$.031
missForest	.379 $\pm$.031	-.257 $\pm$.049	.626 $\pm$.029	.447 $\pm$.031	-.038 $\pm$.057	.610 $\pm$.031	.389 $\pm$.030	-.229 $\pm$.050	.546 $\pm$.030	.475 $\pm$.034	.005 $\pm$.064	.609 $\pm$.034
Sinkhorn	.389 $\pm$.030	-.230 $\pm$.052	.542 $\pm$.037	.448 $\pm$.030	-.025 $\pm$.058	.572 $\pm$.033	.399 $\pm$.030	-.214 $\pm$.052	.539 $\pm$.031	.479 $\pm$.033	.034 $\pm$.064	.560 $\pm$.039
GAIN	.499 $\pm$.027	-.073 $\pm$.053	.482 $\pm$.040	.525 $\pm$.027	.172 $\pm$.049	.585 $\pm$.030	.494 $\pm$.025	-.058 $\pm$.049	.456 $\pm$.035	.575 $\pm$.027	.237 $\pm$.054	.593 $\pm$.035
VAEAC	<u>.565</u> $\pm$.024	.391 $\pm$.028	.693 $\pm$.021	<u>.574</u> $\pm$.025	.421 $\pm$.033	.723 $\pm$.019	.549 $\pm$.025	.339 $\pm$.033	<u>.643</u> $\pm$.026	<u>.587</u> $\pm$.025	.462 $\pm$.032	<u>.693</u> $\pm$.026
MIWAE	.357 $\pm$.031	-.313 $\pm$.048	.669 $\pm$.028	.433 $\pm$.031	-.028 $\pm$.056	.638 $\pm$.028	.376 $\pm$.031	-.299 $\pm$.047	.583 $\pm$.030	.460 $\pm$.034	.032 $\pm$.064	.656 $\pm$.033
not-MIWAE	.355 $\pm$.031	-.290 $\pm$.049	.625 $\pm$.032	.420 $\pm$.031	-.056 $\pm$.057	.583 $\pm$.031	.367 $\pm$.030	-.298 $\pm$.049	.556 $\pm$.032	.456 $\pm$.034	-.020 $\pm$.063	.617 $\pm$.033
MIRACLE	.394 $\pm$.029	-.253 $\pm$.049	.634 $\pm$.022	.448 $\pm$.030	-.089 $\pm$.058	.567 $\pm$.033	.393 $\pm$.028	-.217 $\pm$.051	.521 $\pm$.031	.480 $\pm$.033	.015 $\pm$.059	.612 $\pm$.032
ReMasker	.378 $\pm$.032	-.221 $\pm$.052	.585 $\pm$.032	.441 $\pm$.031	-.018 $\pm$.056	.521 $\pm$.040	.386 $\pm$.030	-.253 $\pm$.051	.509 $\pm$.041	.469 $\pm$.034	.048 $\pm$.061	.586 $\pm$.036
DrIM	.558 $\pm$.024	<u>.269</u> $\pm$.049	.682 $\pm$.027	.584 $\pm$.023	<u>.389</u> $\pm$.049	.674 $\pm$.028	<u>.540</u> $\pm$.023	<u>.267</u> $\pm$.048	.647 $\pm$.027	.616 $\pm$.024	<u>.415</u> $\pm$.047	.703 $\pm$.028

Table 12: Imputation utility at **0.8** missingness rate. The means and their standard errors across 10 datasets and 10 repeated experiments are reported. $\uparrow$ denotes higher is better. The best value is **red bolded**, and the second best is underlined.

Model	MCAR			MAR			MNARL			MNARQ		
	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$
Mean	.322 $\pm$.030	-.180 $\pm$.049	.247 $\pm$.052	.353 $\pm$.031	-.017 $\pm$.056	.278 $\pm$.039	.311 $\pm$.030	-.064 $\pm$.051	.294 $\pm$.043	.413 $\pm$.034	.043 $\pm$.061	.346 $\pm$.041
EM	.366 $\pm$.029	-.186 $\pm$.053	.381 $\pm$.042	.386 $\pm$.029	-.101 $\pm$.060	.418 $\pm$.039	.362 $\pm$.028	-.200 $\pm$.050	.345 $\pm$.039	.441 $\pm$.033	-.039 $\pm$.061	.402 $\pm$.038
SoftImpute	.315 $\pm$.030	-.311 $\pm$.048	.263 $\pm$.048	.362 $\pm$.031	-.077 $\pm$.057	.322 $\pm$.040	.311 $\pm$.030	-.165 $\pm$.050	.312 $\pm$.044	.415 $\pm$.035	.050 $\pm$.060	.376 $\pm$.041
kNNI	.396 $\pm$.031	-.199 $\pm$.055	.585 $\pm$.030	.386 $\pm$.032	-.120 $\pm$.053	<u>.549</u> $\pm$.031	.386 $\pm$.031	-.061 $\pm$.054	<u>.440</u> $\pm$.036	.431 $\pm$.035	-.026 $\pm$.060	.507 $\pm$.039
MICE	.292 $\pm$.031	-.331 $\pm$.049	.476 $\pm$.030	.369 $\pm$.033	-.109 $\pm$.061	.502 $\pm$.035	.295 $\pm$.031	-.260 $\pm$.049	.381 $\pm$.040	.417 $\pm$.036	-.040 $\pm$.060	.494 $\pm$.039
missForest	.315 $\pm$.031	-.306 $\pm$.047	.331 $\pm$.038	.375 $\pm$.032	-.138 $\pm$.055	.426 $\pm$.039	.314 $\pm$.031	-.215 $\pm$.051	.314 $\pm$.042	.422 $\pm$.035	-.030 $\pm$.059	.422 $\pm$.041
Sinkhorn	.330 $\pm$.030	-.211 $\pm$.054	.272 $\pm$.040	.373 $\pm$.031	-.010 $\pm$.055	.389 $\pm$.039	.324 $\pm$.029	-.099 $\pm$.055	.299 $\pm$.039	.425 $\pm$.034	-.063 $\pm$.058	.401 $\pm$.042
GAIN	.411 $\pm$.026	-.161 $\pm$.054	.298 $\pm$.042	.438 $\pm$.027	-.001 $\pm$.054	.364 $\pm$.040	.391 $\pm$.026	-.136 $\pm$.052	.244 $\pm$.040	.514 $\pm$.029	.155 $\pm$.059	.451 $\pm$.041
VAEAC	.505 $\pm$.024	<u>.185</u> $\pm$.039	<u>.534</u> $\pm$.030	.536 $\pm$.025	.316 $\pm$.036	<u>.611</u> $\pm$.026	.492 $\pm$.024	<u>.217</u> $\pm$.041	.488 $\pm$.033	<u>.558</u> $\pm$.026	.322 $\pm$.037	.560 $\pm$.032
MIWAE	.287 $\pm$.031	-.322 $\pm$.049	.397 $\pm$.039	.364 $\pm$.032	-.116 $\pm$.059	.454 $\pm$.036	.299 $\pm$.031	-.283 $\pm$.046	.348 $\pm$.038	.409 $\pm$.035	-.032 $\pm$.060	.487 $\pm$.038
not-MIWAE	.288 $\pm$.030	-.364 $\pm$.042	.357 $\pm$.035	.347 $\pm$.032	-.146 $\pm$.056	.406 $\pm$.039	.288 $\pm$.030	-.332 $\pm$.044	.335 $\pm$.037	.405 $\pm$.035	-.024 $\pm$.057	.473 $\pm$.033
MIRACLE	.320 $\pm$.028	-.273 $\pm$.049	.368 $\pm$.032	.369 $\pm$.030	-.183 $\pm$.058	.367 $\pm$.040	.316 $\pm$.028	-.216 $\pm$.050	.290 $\pm$.036	.419 $\pm$.033	-.097 $\pm$.054	.420 $\pm$.039
ReMasker	.304 $\pm$.028	-.283 $\pm$.053	.299 $\pm$.049	.374 $\pm$.031	-.140 $\pm$.055	.364 $\pm$.042	.306 $\pm$.029	-.285 $\pm$.051	.286 $\pm$.045	.418 $\pm$.035	-.034 $\pm$.058	.418 $\pm$.037
DrIM	<u>.481</u> $\pm$.025	.210 $\pm$.052	.230 $\pm$.038	<u>.519</u> $\pm$.022	<u>.290</u> $\pm$.047	<u>.524</u> $\pm$.034	<u>.471</u> $\pm$.024	.245 $\pm$.051	<u>.262</u> $\pm$.042	.560 $\pm$.025	<u>.290</u> $\pm$.051	<u>.536</u> $\pm$.035

This section evaluates how the imputed datasets perform in downstream machine learning tasks. The analysis in Appendix E.2 focuses on three key metrics:

- **Classification performance (F_1):** The F_1 score is used to assess the balance between precision and recall when classifiers are trained on imputed data. DRIM consistently outperforms other imputation methods in terms of F_1 score. In the experiments, even under varying missingness rates, DRIM maintains a competitive edge in classification accuracy, suggesting that the refined imputation preserves the discriminative features essential for the classifier’s performance.
- **Model selection performance (Model):** The results show that DRIM provides a substantial boost in model selection performance, with significant percentage improvements noted (even up to 300% improvement under severe missingness), especially at higher missingness rates. The improvement is attributed to DRIM’s ability to better capture the underlying data structure, which in turn aids in choosing models that generalize well to unseen data.
- **Feature selection performance (Feature):** DRIM exhibits improved feature selection outcomes by retaining key variables and reducing noise. Although the gains in feature selection are sometimes more modest than those observed in classification and model selection, the overall preservation of the data distribution leads to a more reliable identification of important features.

Overall, the results in Appendix E.2 indicate that by producing imputed data with high fidelity to the original distribution, DRIM substantially enhances the downstream imputation utility. This is reflected in the significant improvements across F_1 scores, model selection, and feature selection metrics.

E.3 Imputation utility for each dataset

We provide the imputation utility performance results for 10 tabular datasets: abalone, anuran, banknote, breast, concrete, kings, letter, loan, redwine, and whitewine, at a 0.3 missingness rate under MCAR, MAR, MNARL, and MNARQ.

Table 13: Imputation utility for each dataset under **MCAR** at 0.3 missingness. The means and their standard errors across 10 repeated experiments are reported. $\uparrow$ denotes higher is better.

model	abalone			anuran			banknote			breast		
	F_1	Model	Feature	F_1	Model	Feature	F_1	Model	Feature	F_1	Model	Feature
Mean	.125 $\pm$.002	-.587 $\pm$.059	.864 $\pm$.015	.858 $\pm$.002	.172 $\pm$.113	.836 $\pm$.012	.863 $\pm$.006	.731 $\pm$.056	1.000 $\pm$.000	.896 $\pm$.006	.198 $\pm$.119	.872 $\pm$.011
kNNI	.135 $\pm$.003	-.475 $\pm$.069	.983 $\pm$.005	.832 $\pm$.003	.105 $\pm$.106	.496 $\pm$.008	.869 $\pm$.006	.641 $\pm$.095	1.000 $\pm$.000	.883 $\pm$.010	-.007 $\pm$.107	.878 $\pm$.014
EM	.128 $\pm$.003	-.505 $\pm$.074	.962 $\pm$.012	.841 $\pm$.003	-.133 $\pm$.091	.457 $\pm$.011	.912 $\pm$.005	.529 $\pm$.075	1.000 $\pm$.000	.889 $\pm$.008	.004 $\pm$.074	.876 $\pm$.015
SoftImpute	.118 $\pm$.004	-.523 $\pm$.054	.967 $\pm$.010	.855 $\pm$.003	.125 $\pm$.126	.626 $\pm$.026	.886 $\pm$.006	.438 $\pm$.055	1.000 $\pm$.000	.881 $\pm$.008	.147 $\pm$.082	.865 $\pm$.015
MICE	.120 $\pm$.003	-.542 $\pm$.048	.974 $\pm$.007	.858 $\pm$.004	.105 $\pm$.111	.545 $\pm$.014	.906 $\pm$.006	.493 $\pm$.056	1.000 $\pm$.000	.888 $\pm$.007	.023 $\pm$.089	.873 $\pm$.016
missForest	.121 $\pm$.003	-.507 $\pm$.045	.976 $\pm$.006	.850 $\pm$.003	.080 $\pm$.109	.485 $\pm$.012	.895 $\pm$.005	.498 $\pm$.047	1.000 $\pm$.000	.880 $\pm$.007	-.072 $\pm$.117	.892 $\pm$.009
Sinkhorn	.125 $\pm$.003	-.547 $\pm$.069	.967 $\pm$.010	.843 $\pm$.003	.015 $\pm$.094	.473 $\pm$.009	.882 $\pm$.004	.483 $\pm$.042	1.000 $\pm$.000	.886 $\pm$.008	.035 $\pm$.111	.885 $\pm$.014
GAIN	.196 $\pm$.003	.019 $\pm$.150	.850 $\pm$.014	.902 $\pm$.011	.452 $\pm$.059	.772 $\pm$.032	.909 $\pm$.012	.577 $\pm$.072	.980 $\pm$.020	.919 $\pm$.008	.113 $\pm$.100	.913 $\pm$.009
VAEAC	.132 $\pm$.002	.322 $\pm$.087	.821 $\pm$.013	.892 $\pm$.000	.535 $\pm$.070	.872 $\pm$.004	.846 $\pm$.003	.623 $\pm$.032	.900 $\pm$.000	.847 $\pm$.004	.655 $\pm$.027	.850 $\pm$.004
MIWAE	.113 $\pm$.003	-.578 $\pm$.047	.967 $\pm$.016	.855 $\pm$.003	.225 $\pm$.099	.517 $\pm$.019	.871 $\pm$.008	.745 $\pm$.060	.980 $\pm$.020	.888 $\pm$.008	-.036 $\pm$.089	.901 $\pm$.008
not-MIWAE	.114 $\pm$.003	-.607 $\pm$.049	.957 $\pm$.011	.854 $\pm$.003	.201 $\pm$.121	.521 $\pm$.014	.867 $\pm$.007	.596 $\pm$.073	1.000 $\pm$.000	.883 $\pm$.009	.017 $\pm$.139	.901 $\pm$.010
MIRACLE	.126 $\pm$.003	-.446 $\pm$.051	.967 $\pm$.009	.821 $\pm$.005	-.188 $\pm$.093	.538 $\pm$.022	.902 $\pm$.007	.525 $\pm$.069	1.000 $\pm$.000	.783 $\pm$.022	-.096 $\pm$.093	.832 $\pm$.020
ReMasker	.119 $\pm$.004	-.543 $\pm$.051	.886 $\pm$.022	.837 $\pm$.003	.045 $\pm$.090	.479 $\pm$.010	.911 $\pm$.006	.497 $\pm$.044	1.000 $\pm$.000	.905 $\pm$.008	.405 $\pm$.125	.849 $\pm$.019
DrIM	.216 $\pm$.003	.827 $\pm$.029	.933 $\pm$.011	.932 $\pm$.009	.569 $\pm$.056	.942 $\pm$.007	.872 $\pm$.007	.718 $\pm$.095	.980 $\pm$.020	.901 $\pm$.008	.430 $\pm$.119	.820 $\pm$.016

model	concrete			kings			letter			loan		
	F_1	Model	Feature	F_1	Model	Feature	F_1	Model	Feature	F_1	Model	Feature
Mean	.326 $\pm$.009	.175 $\pm$.126	.945 $\pm$.007	.440 $\pm$.005	.507 $\pm$.053	.978 $\pm$.002	.584 $\pm$.002	.700 $\pm$.000	.500 $\pm$.015	.916 $\pm$.001	.707 $\pm$.033	.895 $\pm$.007
kNNI	.349 $\pm$.007	.346 $\pm$.156	.940 $\pm$.013	.460 $\pm$.004	.410 $\pm$.064	.972 $\pm$.001	.674 $\pm$.002	.800 $\pm$.000	.948 $\pm$.002	.918 $\pm$.001	.783 $\pm$.047	.902 $\pm$.008
EM	.372 $\pm$.006	.419 $\pm$.210	.952 $\pm$.014	.459 $\pm$.003	.270 $\pm$.047	.945 $\pm$.002	.656 $\pm$.002	.800 $\pm$.000	.959 $\pm$.003	.919 $\pm$.001	.697 $\pm$.037	.878 $\pm$.017
SoftImpute	.328 $\pm$.008	.209 $\pm$.187	.945 $\pm$.010	.445 $\pm$.004	.310 $\pm$.010	.941 $\pm$.004	.594 $\pm$.002	.690 $\pm$.038	.668 $\pm$.015	.915 $\pm$.001	.724 $\pm$.046	.874 $\pm$.008
MICE	.345 $\pm$.008	.341 $\pm$.147	.960 $\pm$.009	.455 $\pm$.004	.290 $\pm$.043	.962 $\pm$.001	.621 $\pm$.002	.860 $\pm$.027	.920 $\pm$.005	.918 $\pm$.001	.733 $\pm$.047	.865 $\pm$.014
missForest	.340 $\pm$.006	.304 $\pm$.130	.952 $\pm$.012	.449 $\pm$.005	.470 $\pm$.058	.946 $\pm$.002	.614 $\pm$.001	.660 $\pm$.031	.742 $\pm$.016	.917 $\pm$.001	.787 $\pm$.054	.858 $\pm$.016
Sinkhorn	.341 $\pm$.006	.513 $\pm$.115	.933 $\pm$.021	.450 $\pm$.004	.490 $\pm$.059	.976 $\pm$.002	.625 $\pm$.001	.620 $\pm$.033	.789 $\pm$.011	.917 $\pm$.001	.646 $\pm$.043	.880 $\pm$.007
GAIN	.430 $\pm$.011	.491 $\pm$.074	.929 $\pm$.011	.491 $\pm$.004	.810 $\pm$.035	.952 $\pm$.004	.599 $\pm$.007	.640 $\pm$.033	.575 $\pm$.022	.915 $\pm$.002	.597 $\pm$.039	.846 $\pm$.010
VAEAC	.363 $\pm$.007	.553 $\pm$.062	.771 $\pm$.015	.513 $\pm$.002	.750 $\pm$.017	.862 $\pm$.002	.712 $\pm$.002	.780 $\pm$.013	.884 $\pm$.002	.827 $\pm$.002	.652 $\pm$.040	.849 $\pm$.011
MIWAE	.294 $\pm$.010	-.095 $\pm$.210	.926 $\pm$.018	.437 $\pm$.005	.450 $\pm$.056	.960 $\pm$.004	.573 $\pm$.003	.812 $\pm$.031	.709 $\pm$.017	.914 $\pm$.001	.720 $\pm$.033	.872 $\pm$.007
not-MIWAE	.313 $\pm$.009	.137 $\pm$.186	.955 $\pm$.014	.441 $\pm$.003	.470 $\pm$.047	.970 $\pm$.003	.599 $\pm$.002	.620 $\pm$.013	.689 $\pm$.013	.916 $\pm$.000	.692 $\pm$.032	.876 $\pm$.010
MIRACLE	.283 $\pm$.011	-.214 $\pm$.177	.924 $\pm$.014	.463 $\pm$.004	.280 $\pm$.055	.946 $\pm$.001	.648 $\pm$.006	.790 $\pm$.023	.940 $\pm$.004	.919 $\pm$.001	.730 $\pm$.054	.882 $\pm$.014
ReMasker	.269 $\pm$.014	.111 $\pm$.111	.921 $\pm$.015	.437 $\pm$.012	.390 $\pm$.108	.929 $\pm$.006	.645 $\pm$.002	.740 $\pm$.031	.931 $\pm$.004	.918 $\pm$.001	.727 $\pm$.040	.887 $\pm$.016
DrIM	.405 $\pm$.007	.388 $\pm$.115	.955 $\pm$.013	.503 $\pm$.009	.940 $\pm$.016	.970 $\pm$.003	.694 $\pm$.004	.950 $\pm$.017	.956 $\pm$.005	.911 $\pm$.001	.647 $\pm$.034	.876 $\pm$.007

model	redwine			whitewine		
	F_1	Model	Feature	F_1	Model	Feature
Mean	.426 $\pm$.007	.206 $\pm$.090	.905 $\pm$.014	.373 $\pm$.005	.086 $\pm$.122	.593 $\pm$.048
kNNI	.440 $\pm$.006	.171 $\pm$.137	.909 $\pm$.021	.386 $\pm$.004	.150 $\pm$.079	.917 $\pm$.008
EM	.435 $\pm$.006	.102 $\pm$.144	.892 $\pm$.017	.384 $\pm$.004	.260 $\pm$.056	.809 $\pm$.023
SoftImpute	.430 $\pm$.007	.047 $\pm$.124	.869 $\pm$.018	.377 $\pm$.005	.160 $\pm$.083	.906 $\pm$.026
MICE	.428 $\pm$.006	-.022 $\pm$.133	.895 $\pm$.020	.388 $\pm$.005	.190 $\pm$.066	.921 $\pm$.021
missForest	.434 $\pm$.005	.001 $\pm$.131	.859 $\pm$.023	.383 $\pm$.004	.070 $\pm$.088	.871 $\pm$.009
Sinkhorn	.435 $\pm$.005	.044 $\pm$.123	.884 $\pm$.018	.386 $\pm$.005	.120 $\pm$.059	.894 $\pm$.013
GAIN	.509 $\pm$.007	.404 $\pm$.110	.897 $\pm$.014	.425 $\pm$.007	.630 $\pm$.072	.846 $\pm$.028
VAEAC	.483 $\pm$.005	.263 $\pm$.046	.785 $\pm$.019	.423 $\pm$.004	.450 $\pm$.017	.722 $\pm$.025
MIWAE	.444 $\pm$.007	-.038 $\pm$.140	.883 $\pm$.017	.380 $\pm$.004	.240 $\pm$.090	.925 $\pm$.007
not-MIWAE	.432 $\pm$.007	.061 $\pm$.151	.909 $\pm$.011	.384 $\pm$.004	.150 $\pm$.092	.914 $\pm$.022
MIRACLE	.430 $\pm$.007	.001 $\pm$.085	.877 $\pm$.020	.379 $\pm$.006	.290 $\pm$.050	.855 $\pm$.018
ReMasker	.415 $\pm$.013	-.030 $\pm$.099	.785 $\pm$.028	.370 $\pm$.008	-.050 $\pm$.120	.563 $\pm$.048
DrIM	.544 $\pm$.007	.535 $\pm$.068	.881 $\pm$.020	.490 $\pm$.004	.366 $\pm$.054	.831 $\pm$.027

Table 14: Imputation utility for each dataset under **MAR** at 0.3 missingness. The means and their standard errors across 10 repeated experiments are reported. $\uparrow$ denotes higher is better.

model	abalone			anuran			banknote			breast		
	F_1	Model	Feature	F_1	Model	Feature	F_1	Model	Feature	F_1	Model	Feature
Mean	.163 $\pm$.014	.147 $\pm$.163	.836 $\pm$.021	.830 $\pm$.018	-.046 $\pm$.150	.552 $\pm$.059	.861 $\pm$.020	.786 $\pm$.065	.800 $\pm$.073	.921 $\pm$.013	.586 $\pm$.098	.810 $\pm$.038
kNNI	.167 $\pm$.013	.142 $\pm$.177	.938 $\pm$.016	.816 $\pm$.019	-.061 $\pm$.149	.683 $\pm$.040	.876 $\pm$.015	.633 $\pm$.074	.940 $\pm$.031	.911 $\pm$.019	.462 $\pm$.124	.898 $\pm$.021
EM	.164 $\pm$.013	.182 $\pm$.140	.902 $\pm$.037	.825 $\pm$.018	-.152 $\pm$.159	.658 $\pm$.044	.884 $\pm$.015	.726 $\pm$.058	.940 $\pm$.031	.909 $\pm$.020	.432 $\pm$.141	.861 $\pm$.024
SoftImpute	.161 $\pm$.012	.211 $\pm$.133	.926 $\pm$.010	.827 $\pm$.018	.005 $\pm$.134	.671 $\pm$.050	.868 $\pm$.020	.673 $\pm$.053	.960 $\pm$.027	.912 $\pm$.018	.536 $\pm$.122	.903 $\pm$.021
MICE	.161 $\pm$.013	.283 $\pm$.149	.962 $\pm$.012	.831 $\pm$.017	-.051 $\pm$.145	.699 $\pm$.046	.881 $\pm$.020	.673 $\pm$.078	.940 $\pm$.031	.910 $\pm$.021	.475 $\pm$.182	.912 $\pm$.018
missForest	.161 $\pm$.013	.313 $\pm$.159	.869 $\pm$.041	.831 $\pm$.017	-.015 $\pm$.157	.650 $\pm$.043	.889 $\pm$.017	.623 $\pm$.093	.960 $\pm$.027	.910 $\pm$.019	.375 $\pm$.181	.910 $\pm$.020
Sinkhorn	.164 $\pm$.013	.044 $\pm$.202	.960 $\pm$.010	.825 $\pm$.018	.055 $\pm$.131	.668 $\pm$.045	.873 $\pm$.020	.658 $\pm$.066	.960 $\pm$.027	.917 $\pm$.015	.497 $\pm$.125	.905 $\pm$.020
GAIN	.206 $\pm$.006	.415 $\pm$.171	.905 $\pm$.015	.863 $\pm$.023	.451 $\pm$.066	.808 $\pm$.051	.934 $\pm$.005	.729 $\pm$.044	1.000 $\pm$.000	.923 $\pm$.013	.674 $\pm$.097	.914 $\pm$.018
VAEAC	.131 $\pm$.005	.497 $\pm$.043	.824 $\pm$.015	.891 $\pm$.001	.565 $\pm$.076	.871 $\pm$.003	.849 $\pm$.004	.678 $\pm$.036	.860 $\pm$.027	.848 $\pm$.003	.622 $\pm$.042	.853 $\pm$.005
MIWAE	.160 $\pm$.013	-.013 $\pm$.194	.931 $\pm$.010	.825 $\pm$.018	-.096 $\pm$.162	.631 $\pm$.045	.865 $\pm$.020	.840 $\pm$.069	.900 $\pm$.033	.908 $\pm$.022	.436 $\pm$.163	.904 $\pm$.023
not-MIWAE	.164 $\pm$.013	.055 $\pm$.157	.914 $\pm$.019	.825 $\pm$.018	-.065 $\pm$.147	.681 $\pm$.043	.863 $\pm$.020	.570 $\pm$.097	.880 $\pm$.033	.909 $\pm$.019	.550 $\pm$.150	.895 $\pm$.024
MIRACLE	.163 $\pm$.013	.146 $\pm$.146	.900 $\pm$.035	.822 $\pm$.018	-.113 $\pm$.144	.701 $\pm$.046	.889 $\pm$.015	.709 $\pm$.055	.960 $\pm$.027	.883 $\pm$.021	.250 $\pm$.169	.770 $\pm$.027
ReMasker	.169 $\pm$.012	.283 $\pm$.173	.888 $\pm$.043	.815 $\pm$.019	-.183 $\pm$.170	.668 $\pm$.041	.887 $\pm$.018	.740 $\pm$.045	.940 $\pm$.031	.905 $\pm$.017	.316 $\pm$.183	.816 $\pm$.022
DrIM	.211 $\pm$.007	.761 $\pm$.059	.883 $\pm$.024	.937 $\pm$.008	.300 $\pm$.127	.925 $\pm$.011	.877 $\pm$.020	.607 $\pm$.090	.920 $\pm$.033	.894 $\pm$.022	.518 $\pm$.134	.840 $\pm$.037

model	concrete			kings			letter			loan		
	F_1	Model	Feature	F_1	Model	Feature	F_1	Model	Feature	F_1	Model	Feature
Mean	.350 $\pm$.025	.511 $\pm$.100	.881 $\pm$.028	.421 $\pm$.009	.678 $\pm$.083	.900 $\pm$.017	.621 $\pm$.028	.840 $\pm$.050	.715 $\pm$.032	.915 $\pm$.003	.670 $\pm$.063	.898 $\pm$.020
kNNI	.372 $\pm$.024	.769 $\pm$.098	.929 $\pm$.013	.432 $\pm$.009	.656 $\pm$.084	.957 $\pm$.005	.664 $\pm$.028	.832 $\pm$.029	.936 $\pm$.012	.919 $\pm$.002	.804 $\pm$.047	.935 $\pm$.019
EM	.391 $\pm$.024	.733 $\pm$.043	.845 $\pm$.037	.423 $\pm$.010	.689 $\pm$.065	.913 $\pm$.012	.666 $\pm$.025	.840 $\pm$.027	.952 $\pm$.006	.893 $\pm$.022	.732 $\pm$.048	.910 $\pm$.021
SoftImpute	.349 $\pm$.023	.620 $\pm$.101	.879 $\pm$.026	.419 $\pm$.010	.600 $\pm$.091	.938 $\pm$.006	.627 $\pm$.029	.880 $\pm$.025	.781 $\pm$.023	.916 $\pm$.003	.673 $\pm$.042	.910 $\pm$.021
MICE	.370 $\pm$.027	.549 $\pm$.132	.886 $\pm$.029	.423 $\pm$.010	.562 $\pm$.089	.942 $\pm$.007	.646 $\pm$.028	.890 $\pm$.023	.894 $\pm$.014	.917 $\pm$.002	.759 $\pm$.052	.908 $\pm$.027
missForest	.353 $\pm$.022	.601 $\pm$.149	.826 $\pm$.033	.428 $\pm$.010	.700 $\pm$.087	.948 $\pm$.005	.637 $\pm$.027	.857 $\pm$.040	.803 $\pm$.027	.918 $\pm$.002	.761 $\pm$.045	.902 $\pm$.026
Sinkhorn	.370 $\pm$.023	.595 $\pm$.164	.860 $\pm$.020	.423 $\pm$.009	.725 $\pm$.070	.966 $\pm$.004	.646 $\pm$.027	.840 $\pm$.045	.843 $\pm$.020	.915 $\pm$.003	.731 $\pm$.039	.912 $\pm$.021
GAIN	.421 $\pm$.013	.624 $\pm$.080	.869 $\pm$.022	.500 $\pm$.011	.760 $\pm$.048	.959 $\pm$.008	.645 $\pm$.027	.810 $\pm$.038	.794 $\pm$.017	.915 $\pm$.003	.572 $\pm$.059	.888 $\pm$.020
VAEAC	.357 $\pm$.007	.649 $\pm$.051	.721 $\pm$.027	.514 $\pm$.001	.770 $\pm$.015	.860 $\pm$.006	.724 $\pm$.002	.800 $\pm$.000	.890 $\pm$.002	.827 $\pm$.001	.640 $\pm$.054	.821 $\pm$.020
MIWAE	.335 $\pm$.023	.389 $\pm$.111	.845 $\pm$.038	.416 $\pm$.010	.644 $\pm$.078	.949 $\pm$.006	.616 $\pm$.029	.880 $\pm$.025	.809 $\pm$.010	.916 $\pm$.003	.802 $\pm$.052	.891 $\pm$.023
not-MIWAE	.338 $\pm$.023	.602 $\pm$.089	.879 $\pm$.026	.423 $\pm$.008	.690 $\pm$.078	.949 $\pm$.006	.629 $\pm$.028	.897 $\pm$.021	.805 $\pm$.020	.916 $\pm$.003	.744 $\pm$.048	.908 $\pm$.021
MIRACLE	.339 $\pm$.025	.247 $\pm$.147	.893 $\pm$.019	.436 $\pm$.009	.578 $\pm$.074	.950 $\pm$.010	.671 $\pm$.027	.840 $\pm$.027	.951 $\pm$.009	.920 $\pm$.002	.834 $\pm$.040	.931 $\pm$.025
ReMasker	.332 $\pm$.025	.263 $\pm$.084	.893 $\pm$.019	.423 $\pm$.014	.533 $\pm$.078	.935 $\pm$.006	.659 $\pm$.027	.840 $\pm$.027	.947 $\pm$.008	.919 $\pm$.002	.803 $\pm$.048	.910 $\pm$.026
DrIM	.412 $\pm$.010	.545 $\pm$.112	.898 $\pm$.015	.556 $\pm$.004	.950 $\pm$.017	.964 $\pm$.004	.775 $\pm$.008	1.000 $\pm$.000	.972 $\pm$.005	.915 $\pm$.003	.684 $\pm$.071	.898 $\pm$.022

model	redwine			whitewine		
	F_1	Model	Feature	F_1	Model	Feature
Mean	.428 $\pm$.027	.673 $\pm$.092	.543 $\pm$.085	.386 $\pm$.021	.647 $\pm$.096	.525 $\pm$.071
kNNI	.437 $\pm$.027	.697 $\pm$.115	.716 $\pm$.064	.395 $\pm$.021	.675 $\pm$.044	.757 $\pm$.056
EM	.436 $\pm$.024	.502 $\pm$.122	.699 $\pm$.053	.398 $\pm$.022	.660 $\pm$.096	.667 $\pm$.078
SoftImpute	.421 $\pm$.026	.571 $\pm$.116	.689 $\pm$.067	.387 $\pm$.023	.777 $\pm$.078	.806 $\pm$.038
MICE	.430 $\pm$.026	.633 $\pm$.103	.686 $\pm$.058	.390 $\pm$.024	.660 $\pm$.081	.769 $\pm$.054
missForest	.426 $\pm$.026	.597 $\pm$.108	.717 $\pm$.061	.387 $\pm$.023	.687 $\pm$.093	.753 $\pm$.049
Sinkhorn	.431 $\pm$.025	.679 $\pm$.109	.686 $\pm$.056	.389 $\pm$.022	.660 $\pm$.086	.714 $\pm$.043
GAIN	.526 $\pm$.009	.382 $\pm$.078	.812 $\pm$.028	.467 $\pm$.013	.600 $\pm$.074	.822 $\pm$.062
VAEAC	.485 $\pm$.004	.403 $\pm$.053	.808 $\pm$.018	.431 $\pm$.002	.500 $\pm$.037	.757 $\pm$.025
MIWAE	.431 $\pm$.026	.602 $\pm$.094	.676 $\pm$.057	.389 $\pm$.023	.720 $\pm$.074	.814 $\pm$.035
not-MIWAE	.420 $\pm$.027	.622 $\pm$.118	.704 $\pm$.057	.387 $\pm$.022	.740 $\pm$.048	.773 $\pm$.044
MIRACLE	.435 $\pm$.024	.487 $\pm$.125	.731 $\pm$.057	.393 $\pm$.022	.760 $\pm$.070	.731 $\pm$.045
ReMasker	.415 $\pm$.029	.603 $\pm$.092	.567 $\pm$.074	.381 $\pm$.022	.605 $\pm$.104	.615 $\pm$.086
DrIM	.549 $\pm$.008	.476 $\pm$.083	.914 $\pm$.010	.498 $\pm$.004	.540 $\pm$.065	.810 $\pm$.040

Table 15: Imputation utility for each dataset under **MNARL** at 0.3 missingness. The means and their standard errors across 10 repeated experiments are reported. $\uparrow$ denotes higher is better.

model	abalone			anuran			banknote			breast		
	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$
Mean	.131 $\pm$.005	-.206 $\pm$.080	.895 $\pm$.018	.817 $\pm$.010	.035 $\pm$.100	.683 $\pm$.039	.839 $\pm$.015	.677 $\pm$.084	.860 $\pm$.060	.903 $\pm$.009	.529 $\pm$.116	.783 $\pm$.027
kNNI	.140 $\pm$.004	-.220 $\pm$.133	.924 $\pm$.017	.788 $\pm$.007	-.233 $\pm$.126	.513 $\pm$.029	.846 $\pm$.014	.701 $\pm$.089	.940 $\pm$.031	.890 $\pm$.014	.175 $\pm$.115	.866 $\pm$.021
EM	.133 $\pm$.005	-.238 $\pm$.125	.926 $\pm$.013	.801 $\pm$.006	-.313 $\pm$.113	.519 $\pm$.038	.886 $\pm$.013	.690 $\pm$.040	.940 $\pm$.031	.889 $\pm$.017	.047 $\pm$.127	.861 $\pm$.023
SoftImpute	.125 $\pm$.005	-.297 $\pm$.137	.914 $\pm$.016	.811 $\pm$.007	-.090 $\pm$.123	.576 $\pm$.047	.859 $\pm$.014	.503 $\pm$.060	.920 $\pm$.033	.890 $\pm$.015	.185 $\pm$.123	.849 $\pm$.022
MICE	.126 $\pm$.005	-.457 $\pm$.070	.955 $\pm$.013	.813 $\pm$.007	-.237 $\pm$.131	.577 $\pm$.044	.879 $\pm$.016	.627 $\pm$.060	.940 $\pm$.031	.889 $\pm$.018	.161 $\pm$.139	.865 $\pm$.024
missForest	.129 $\pm$.005	-.282 $\pm$.104	.893 $\pm$.023	.810 $\pm$.007	-.160 $\pm$.139	.518 $\pm$.032	.879 $\pm$.014	.583 $\pm$.093	.940 $\pm$.031	.889 $\pm$.016	.119 $\pm$.156	.861 $\pm$.026
Sinkhorn	.133 $\pm$.004	-.374 $\pm$.093	.940 $\pm$.010	.803 $\pm$.008	-.172 $\pm$.125	.505 $\pm$.032	.857 $\pm$.014	.658 $\pm$.073	.940 $\pm$.031	.898 $\pm$.012	.201 $\pm$.138	.871 $\pm$.022
GAIN	.187 $\pm$.005	-.008 $\pm$.180	.898 $\pm$.027	.875 $\pm$.020	.088 $\pm$.117	.751 $\pm$.043	.901 $\pm$.011	.587 $\pm$.059	.980 $\pm$.020	.922 $\pm$.006	.525 $\pm$.158	.865 $\pm$.031
VAEAC	.109 $\pm$.023	.369 $\pm$.081	.740 $\pm$.095	.890 $\pm$.001	.538 $\pm$.081	.870 $\pm$.004	.845 $\pm$.003	.612 $\pm$.047	.900 $\pm$.000	.843 $\pm$.003	.644 $\pm$.047	.841 $\pm$.004
MIWAE	.121 $\pm$.005	-.417 $\pm$.074	.921 $\pm$.017	.811 $\pm$.007	-.133 $\pm$.142	.522 $\pm$.032	.848 $\pm$.016	.713 $\pm$.077	.940 $\pm$.031	.889 $\pm$.018	.088 $\pm$.165	.872 $\pm$.020
not-MIWAE	.126 $\pm$.005	-.387 $\pm$.100	.933 $\pm$.019	.809 $\pm$.008	-.141 $\pm$.156	.547 $\pm$.038	.840 $\pm$.014	.685 $\pm$.046	1.000 $\pm$.000	.891 $\pm$.017	.162 $\pm$.149	.879 $\pm$.020
MIRACLE	.129 $\pm$.005	-.276 $\pm$.076	.914 $\pm$.012	.787 $\pm$.009	-.304 $\pm$.104	.589 $\pm$.035	.878 $\pm$.014	.577 $\pm$.050	.920 $\pm$.061	.818 $\pm$.017	-.140 $\pm$.162	.755 $\pm$.033
ReMasker	.122 $\pm$.006	-.417 $\pm$.090	.910 $\pm$.025	.789 $\pm$.007	-.283 $\pm$.135	.494 $\pm$.036	.874 $\pm$.014	.580 $\pm$.094	.960 $\pm$.027	.901 $\pm$.007	.274 $\pm$.113	.759 $\pm$.035
DrIM	.214 $\pm$.003	.754 $\pm$.048	.945 $\pm$.011	.944 $\pm$.007	.206 $\pm$.133	.931 $\pm$.008	.879 $\pm$.006	.433 $\pm$.154	.960 $\pm$.027	.865 $\pm$.018	.299 $\pm$.167	.764 $\pm$.036

model	concrete			kings			letter			loan		
	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$
Mean	.307 $\pm$.013	.184 $\pm$.110	.917 $\pm$.020	.413 $\pm$.010	.730 $\pm$.065	.959 $\pm$.005	.561 $\pm$.005	.740 $\pm$.048	.665 $\pm$.030	.912 $\pm$.002	.630 $\pm$.065	.881 $\pm$.023
kNNI	.328 $\pm$.009	.534 $\pm$.093	.869 $\pm$.028	.432 $\pm$.010	.640 $\pm$.095	.956 $\pm$.003	.651 $\pm$.007	.800 $\pm$.000	.950 $\pm$.005	.916 $\pm$.002	.740 $\pm$.027	.902 $\pm$.020
EM	.359 $\pm$.013	.654 $\pm$.080	.707 $\pm$.070	.432 $\pm$.009	.540 $\pm$.083	.926 $\pm$.010	.632 $\pm$.004	.810 $\pm$.028	.953 $\pm$.005	.916 $\pm$.001	.697 $\pm$.038	.855 $\pm$.020
SoftImpute	.307 $\pm$.013	.184 $\pm$.110	.917 $\pm$.020	.416 $\pm$.009	.500 $\pm$.099	.926 $\pm$.004	.569 $\pm$.005	.777 $\pm$.051	.749 $\pm$.026	.913 $\pm$.002	.702 $\pm$.040	.868 $\pm$.014
MICE	.322 $\pm$.012	.433 $\pm$.104	.740 $\pm$.075	.430 $\pm$.009	.390 $\pm$.075	.942 $\pm$.004	.595 $\pm$.006	.840 $\pm$.027	.906 $\pm$.012	.917 $\pm$.002	.714 $\pm$.040	.859 $\pm$.021
missForest	.317 $\pm$.010	.227 $\pm$.090	.788 $\pm$.055	.423 $\pm$.009	.680 $\pm$.074	.936 $\pm$.004	.586 $\pm$.005	.740 $\pm$.052	.796 $\pm$.022	.914 $\pm$.002	.750 $\pm$.052	.876 $\pm$.018
Sinkhorn	.324 $\pm$.010	.393 $\pm$.139	.762 $\pm$.061	.422 $\pm$.008	.730 $\pm$.065	.962 $\pm$.003	.599 $\pm$.006	.730 $\pm$.050	.856 $\pm$.016	.912 $\pm$.003	.688 $\pm$.052	.884 $\pm$.019
GAIN	.414 $\pm$.009	.371 $\pm$.153	.900 $\pm$.017	.488 $\pm$.008	.770 $\pm$.037	.963 $\pm$.007	.580 $\pm$.010	.620 $\pm$.042	.690 $\pm$.037	.911 $\pm$.002	.587 $\pm$.015	.890 $\pm$.021
VAEAC	.347 $\pm$.010	.472 $\pm$.056	.743 $\pm$.036	.510 $\pm$.002	.760 $\pm$.016	.854 $\pm$.002	.712 $\pm$.002	.800 $\pm$.000	.887 $\pm$.002	.825 $\pm$.001	.620 $\pm$.042	.838 $\pm$.014
MIWAE	.276 $\pm$.012	.095 $\pm$.167	.843 $\pm$.041	.415 $\pm$.008	.730 $\pm$.079	.949 $\pm$.002	.549 $\pm$.005	.860 $\pm$.022	.772 $\pm$.025	.914 $\pm$.002	.713 $\pm$.052	.858 $\pm$.022
not-MIWAE	.290 $\pm$.010	.441 $\pm$.145	.831 $\pm$.041	.410 $\pm$.010	.682 $\pm$.072	.947 $\pm$.003	.573 $\pm$.005	.740 $\pm$.050	.760 $\pm$.028	.912 $\pm$.002	.710 $\pm$.035	.869 $\pm$.019
MIRACLE	.289 $\pm$.019	-.064 $\pm$.121	.860 $\pm$.032	.437 $\pm$.007	.460 $\pm$.095	.941 $\pm$.005	.620 $\pm$.009	.820 $\pm$.013	.932 $\pm$.007	.918 $\pm$.002	.783 $\pm$.040	.878 $\pm$.022
ReMasker	.273 $\pm$.018	.141 $\pm$.141	.905 $\pm$.019	.428 $\pm$.012	.430 $\pm$.092	.905 $\pm$.011	.624 $\pm$.005	.780 $\pm$.020	.942 $\pm$.007	.918 $\pm$.002	.760 $\pm$.041	.882 $\pm$.024
DrIM	.386 $\pm$.008	.356 $\pm$.085	.931 $\pm$.016	.517 $\pm$.010	.940 $\pm$.016	.965 $\pm$.003	.677 $\pm$.010	.930 $\pm$.015	.942 $\pm$.004	.909 $\pm$.003	.677 $\pm$.036	.857 $\pm$.013

model	redwine			whitewine		
	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$
Mean	.375 $\pm$.009	.422 $\pm$.114	.682 $\pm$.072	.344 $\pm$.007	.540 $\pm$.117	.648 $\pm$.069
kNNI	.400 $\pm$.010	.477 $\pm$.161	.705 $\pm$.049	.357 $\pm$.008	.577 $\pm$.119	.804 $\pm$.047
EM	.395 $\pm$.009	.529 $\pm$.121	.666 $\pm$.052	.360 $\pm$.008	.656 $\pm$.097	.688 $\pm$.070
SoftImpute	.386 $\pm$.010	.426 $\pm$.113	.671 $\pm$.050	.342 $\pm$.007	.530 $\pm$.125	.792 $\pm$.044
MICE	.390 $\pm$.012	.467 $\pm$.148	.677 $\pm$.052	.349 $\pm$.009	.450 $\pm$.138	.788 $\pm$.062
missForest	.390 $\pm$.011	.402 $\pm$.142	.712 $\pm$.050	.350 $\pm$.007	.606 $\pm$.109	.773 $\pm$.055
Sinkhorn	.389 $\pm$.011	.454 $\pm$.135	.720 $\pm$.052	.352 $\pm$.006	.465 $\pm$.143	.774 $\pm$.059
GAIN	.506 $\pm$.007	.382 $\pm$.110	.796 $\pm$.031	.427 $\pm$.010	.537 $\pm$.075	.695 $\pm$.095
VAEAC	.472 $\pm$.005	.266 $\pm$.115	.785 $\pm$.016	.423 $\pm$.004	.450 $\pm$.031	.700 $\pm$.021
MIWAE	.392 $\pm$.010	.506 $\pm$.121	.715 $\pm$.060	.347 $\pm$.008	.537 $\pm$.130	.817 $\pm$.046
not-MIWAE	.383 $\pm$.011	.479 $\pm$.116	.662 $\pm$.055	.346 $\pm$.007	.575 $\pm$.115	.755 $\pm$.050
MIRACLE	.399 $\pm$.009	.309 $\pm$.145	.699 $\pm$.056	.360 $\pm$.005	.470 $\pm$.114	.765 $\pm$.058
ReMasker	.388 $\pm$.020	.185 $\pm$.098	.534 $\pm$.056	.339 $\pm$.010	.342 $\pm$.169	.532 $\pm$.076
DrIM	.542 $\pm$.008	.233 $\pm$.040	.858 $\pm$.020	.488 $\pm$.005	.440 $\pm$.078	.839 $\pm$.039

Table 16: Imputation utility for each dataset under **MNARQ** at 0.3 missingness. The means and their standard errors across 10 repeated experiments are reported. $\uparrow$ denotes higher is better.

model	abalone			anuran			banknote			breast		
	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$
Mean	.143 $\pm$.015	-.292 $\pm$.211	.774 $\pm$.035	.921 $\pm$.021	.278 $\pm$.118	.549 $\pm$.075	.904 $\pm$.012	.832 $\pm$.053	.800 $\pm$.052	.915 $\pm$.011	.469 $\pm$.164	.841 $\pm$.032
kNNI	.146 $\pm$.013	-.284 $\pm$.198	.926 $\pm$.021	.914 $\pm$.023	.333 $\pm$.128	.700 $\pm$.052	.916 $\pm$.011	.680 $\pm$.075	1.000 $\pm$.000	.915 $\pm$.011	.513 $\pm$.176	.923 $\pm$.016
EM	.142 $\pm$.013	-.317 $\pm$.173	.855 $\pm$.033	.911 $\pm$.021	.223 $\pm$.120	.648 $\pm$.052	.922 $\pm$.007	.753 $\pm$.071	.960 $\pm$.027	.915 $\pm$.012	.333 $\pm$.205	.884 $\pm$.019
SoftImpute	.141 $\pm$.014	-.262 $\pm$.199	.924 $\pm$.023	.918 $\pm$.022	.300 $\pm$.118	.717 $\pm$.063	.911 $\pm$.011	.604 $\pm$.087	1.000 $\pm$.000	.915 $\pm$.010	.406 $\pm$.202	.915 $\pm$.018
MICE	.143 $\pm$.015	-.213 $\pm$.194	.950 $\pm$.017	.920 $\pm$.021	.344 $\pm$.120	.738 $\pm$.052	.922 $\pm$.010	.711 $\pm$.070	1.000 $\pm$.000	.918 $\pm$.013	.231 $\pm$.216	.906 $\pm$.025
missForest	.145 $\pm$.014	-.262 $\pm$.205	.919 $\pm$.024	.919 $\pm$.022	.325 $\pm$.125	.687 $\pm$.050	.920 $\pm$.010	.600 $\pm$.066	1.000 $\pm$.000	.915 $\pm$.012	.380 $\pm$.175	.927 $\pm$.013
Sinkhorn	.143 $\pm$.013	-.154 $\pm$.172	.929 $\pm$.020	.917 $\pm$.022	.316 $\pm$.121	.702 $\pm$.053	.912 $\pm$.011	.670 $\pm$.073	1.000 $\pm$.000	.920 $\pm$.011	.409 $\pm$.195	.919 $\pm$.022
GAIN	.182 $\pm$.012	.021 $\pm$.198	.907 $\pm$.019	.951 $\pm$.014	.490 $\pm$.100	.831 $\pm$.028	.943 $\pm$.004	.720 $\pm$.043	1.000 $\pm$.000	.929 $\pm$.009	.424 $\pm$.161	.925 $\pm$.020
VAEAC	.136 $\pm$.004	.538 $\pm$.049	.829 $\pm$.011	.893 $\pm$.001	.613 $\pm$.082	.871 $\pm$.003	.854 $\pm$.003	.625 $\pm$.035	.900 $\pm$.000	.850 $\pm$.003	.624 $\pm$.044	.850 $\pm$.005
MIWAE	.138 $\pm$.016	-.307 $\pm$.202	.948 $\pm$.010	.921 $\pm$.021	.364 $\pm$.128	.700 $\pm$.058	.904 $\pm$.013	.774 $\pm$.050	1.000 $\pm$.000	.919 $\pm$.011	.412 $\pm$.195	.900 $\pm$.027
not-MIWAE	.141 $\pm$.014	-.217 $\pm$.197	.950 $\pm$.015	.919 $\pm$.022	.356 $\pm$.122	.726 $\pm$.053	.902 $\pm$.012	.819 $\pm$.045	.980 $\pm$.020	.915 $\pm$.012	.474 $\pm$.171	.912 $\pm$.021
MIRACLE	.146 $\pm$.014	-.224 $\pm$.185	.924 $\pm$.025	.917 $\pm$.022	.288 $\pm$.126	.726 $\pm$.052	.919 $\pm$.013	.739 $\pm$.056	1.000 $\pm$.000	.894 $\pm$.017	.088 $\pm$.174	.788 $\pm$.038
ReMasker	.148 $\pm$.013	-.197 $\pm$.154	.798 $\pm$.037	.913 $\pm$.024	.267 $\pm$.112	.697 $\pm$.050	.918 $\pm$.010	.663 $\pm$.076	1.000 $\pm$.000	.911 $\pm$.009	.194 $\pm$.181	.846 $\pm$.028
DrIM	.210 $\pm$.005	.787 $\pm$.049	.876 $\pm$.022	.969 $\pm$.008	.666 $\pm$.074	.937 $\pm$.011	.912 $\pm$.011	.793 $\pm$.045	.960 $\pm$.027	.913 $\pm$.011	.527 $\pm$.150	.886 $\pm$.021

model	concrete			kings			letter			loan		
	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$
Mean	.351 $\pm$.021	.223 $\pm$.175	.945 $\pm$.009	.521 $\pm$.024	.610 $\pm$.119	.952 $\pm$.006	.646 $\pm$.025	.800 $\pm$.042	.651 $\pm$.048	.921 $\pm$.002	.829 $\pm$.048	.926 $\pm$.008
kNNI	.370 $\pm$.020	.634 $\pm$.091	.819 $\pm$.032	.535 $\pm$.025	.580 $\pm$.125	.976 $\pm$.001	.680 $\pm$.025	.720 $\pm$.053	.911 $\pm$.014	.923 $\pm$.001	.867 $\pm$.047	.947 $\pm$.010
EM	.381 $\pm$.019	.562 $\pm$.112	.660 $\pm$.067	.506 $\pm$.027	.529 $\pm$.122	.965 $\pm$.006	.687 $\pm$.022	.800 $\pm$.042	.951 $\pm$.004	.893 $\pm$.024	.847 $\pm$.051	.927 $\pm$.007
SoftImpute	.351 $\pm$.020	.386 $\pm$.123	.807 $\pm$.057	.525 $\pm$.025	.580 $\pm$.131	.962 $\pm$.003	.652 $\pm$.025	.780 $\pm$.049	.764 $\pm$.028	.922 $\pm$.001	.847 $\pm$.049	.933 $\pm$.007
MICE	.367 $\pm$.022	.418 $\pm$.152	.717 $\pm$.067	.532 $\pm$.024	.535 $\pm$.124	.961 $\pm$.006	.668 $\pm$.024	.850 $\pm$.037	.913 $\pm$.007	.922 $\pm$.002	.852 $\pm$.037	.944 $\pm$.009
missForest	.360 $\pm$.021	.344 $\pm$.134	.705 $\pm$.069	.526 $\pm$.024	.660 $\pm$.109	.963 $\pm$.005	.663 $\pm$.024	.780 $\pm$.049	.806 $\pm$.026	.923 $\pm$.001	.770 $\pm$.045	.919 $\pm$.013
Sinkhorn	.360 $\pm$.021	.639 $\pm$.094	.638 $\pm$.074	.529 $\pm$.024	.630 $\pm$.122	.982 $\pm$.001	.670 $\pm$.024	.720 $\pm$.053	.819 $\pm$.021	.922 $\pm$.002	.783 $\pm$.047	.938 $\pm$.010
GAIN	.435 $\pm$.012	.623 $\pm$.051	.731 $\pm$.072	.558 $\pm$.018	.830 $\pm$.038	.973 $\pm$.005	.680 $\pm$.022	.740 $\pm$.047	.764 $\pm$.030	.919 $\pm$.002	.660 $\pm$.064	.922 $\pm$.017
VAEAC	.370 $\pm$.011	.552 $\pm$.065	.557 $\pm$.069	.519 $\pm$.002	.760 $\pm$.016	.872 $\pm$.004	.727 $\pm$.002	.800 $\pm$.000	.891 $\pm$.002	.828 $\pm$.002	.627 $\pm$.046	.850 $\pm$.015
MIWAE	.330 $\pm$.021	.170 $\pm$.154	.871 $\pm$.029	.520 $\pm$.025	.600 $\pm$.134	.967 $\pm$.002	.641 $\pm$.025	.890 $\pm$.028	.802 $\pm$.019	.920 $\pm$.001	.835 $\pm$.046	.890 $\pm$.010
not-MIWAE	.350 $\pm$.020	.334 $\pm$.155	.838 $\pm$.040	.527 $\pm$.025	.630 $\pm$.122	.974 $\pm$.003	.657 $\pm$.025	.780 $\pm$.051	.781 $\pm$.026	.922 $\pm$.002	.807 $\pm$.052	.937 $\pm$.009
MIRACLE	.336 $\pm$.020	-.174 $\pm$.132	.829 $\pm$.044	.536 $\pm$.025	.550 $\pm$.123	.968 $\pm$.006	.691 $\pm$.024	.820 $\pm$.036	.941 $\pm$.008	.923 $\pm$.001	.820 $\pm$.043	.939 $\pm$.010
ReMasker	.309 $\pm$.021	.146 $\pm$.143	.886 $\pm$.020	.518 $\pm$.022	.620 $\pm$.119	.949 $\pm$.007	.677 $\pm$.025	.760 $\pm$.050	.931 $\pm$.010	.923 $\pm$.001	.787 $\pm$.037	.930 $\pm$.013
DrIM	.412 $\pm$.010	.586 $\pm$.062	.845 $\pm$.038	.588 $\pm$.006	.950 $\pm$.017	.964 $\pm$.006	.789 $\pm$.006	1.000 $\pm$.000	.979 $\pm$.002	.920 $\pm$.002	.820 $\pm$.055	.926 $\pm$.014

model	redwine			whitewine		
	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$	$F_1 \uparrow$	Model $\uparrow$	Feature $\uparrow$
Mean	.466 $\pm$.018	.130 $\pm$.127	.787 $\pm$.036	.416 $\pm$.018	.290 $\pm$.097	.485 $\pm$.080
kNNI	.469 $\pm$.020	.145 $\pm$.114	.843 $\pm$.024	.423 $\pm$.017	.295 $\pm$.090	.730 $\pm$.056
EM	.469 $\pm$.021	.083 $\pm$.126	.859 $\pm$.025	.420 $\pm$.019	.320 $\pm$.083	.655 $\pm$.086
SoftImpute	.467 $\pm$.020	.060 $\pm$.149	.874 $\pm$.023	.416 $\pm$.018	.330 $\pm$.082	.811 $\pm$.040
MICE	.470 $\pm$.020	.104 $\pm$.137	.858 $\pm$.023	.417 $\pm$.019	.270 $\pm$.068	.777 $\pm$.049
missForest	.465 $\pm$.021	.129 $\pm$.135	.879 $\pm$.022	.417 $\pm$.018	.280 $\pm$.083	.770 $\pm$.038
Sinkhorn	.468 $\pm$.020	.170 $\pm$.110	.838 $\pm$.020	.419 $\pm$.019	.300 $\pm$.095	.639 $\pm$.103
GAIN	.529 $\pm$.009	.458 $\pm$.088	.895 $\pm$.012	.460 $\pm$.013	.710 $\pm$.048	.800 $\pm$.040
VAEAC	.490 $\pm$.005	.285 $\pm$.062	.804 $\pm$.009	.435 $\pm$.004	.510 $\pm$.043	.803 $\pm$.013
MIWAE	.463 $\pm$.021	.005 $\pm$.139	.813 $\pm$.037	.420 $\pm$.018	.305 $\pm$.081	.741 $\pm$.067
not-MIWAE	.463 $\pm$.019	.062 $\pm$.125	.875 $\pm$.017	.419 $\pm$.018	.330 $\pm$.090	.770 $\pm$.067
MIRACLE	.467 $\pm$.023	.086 $\pm$.151	.864 $\pm$.030	.420 $\pm$.019	.340 $\pm$.093	.671 $\pm$.087
ReMasker	.449 $\pm$.023	.039 $\pm$.138	.745 $\pm$.033	.405 $\pm$.019	.117 $\pm$.148	.572 $\pm$.054
DrIM	.559 $\pm$.005	.466 $\pm$.075	.904 $\pm$.018	.503 $\pm$.003	.626 $\pm$.048	.853 $\pm$.024

In Appendix E.3, the experimental results are broken down on a per-dataset basis, providing insights into the performance consistency of DRIM across diverse real-world tabular datasets. The analysis is structured as follows:

- **Dataset-level performance consistency:** The reported results cover 10 distinct datasets at 0.3 missingness rate. DRIM demonstrates a consistent advantage across nearly all datasets, which confirms that its imputation strategy is robust and adaptable to various data types and distributions. This consistency is crucial for practical applications where dataset characteristics can vary widely.
- **Generalization:** The datasets differ in size, feature composition, and inherent statistical properties. DRIM’s strong performance across these diverse datasets highlights its superior generalization capability. Each dataset analysis reveals that even in cases where baseline methods perform adequately on one dataset, DRIM either matches or surpasses their performance on others, underscoring its ability to generalize well across different data domains.

These dataset-specific results from Appendix E.3 confirm that the benefits of DRIM are not limited to an aggregated measure over all datasets but are evident even when scrutinized at the individual dataset level.

E.4 Effect of contrastive learning

In this section, we present the results on the effect of contrastive learning on DRIM’s imputation performance under MCAR, MNARL, and MNARQ.

Table 17: **Effect of contrastive learning** under MCAR. ‘*w/o*’ refers to using a pre-trained BERT model without fine-tuning. Δ indicates the performance difference between the two settings, and the **bold** highlights the positive improvement. The means across 10 datasets and 10 repeated experiments are reported. $\uparrow$ ($\downarrow$) denotes that higher (lower) is better.

rate	ASMAPE $\downarrow$			KL $\downarrow$			$F_1 \uparrow$			Model $\uparrow$			Feature $\uparrow$		
	<i>w/o</i>	DRIM	Δ	<i>w/o</i>	DRIM	Δ	<i>w/o</i>	DRIM	Δ	<i>w/o</i>	DRIM	Δ	<i>w/o</i>	DRIM	Δ
0.2	.841	.839	+0.24%	.022	.018	+18.18%	.665	.670	+0.75%	.692	.719	+3.90%	.949	.945	−0.42%
0.3	.849	.845	+0.47%	.038	.033	+13.16%	.638	.647	+1.41%	.517	.637	+23.21%	.920	.914	−0.65%
0.4	.863	.863	0.00%	.060	.054	+10.00%	.602	.620	+2.99%	.301	.493	+63.79%	.853	.871	+2.11%
0.6	.888	.875	+1.46%	.119	.101	+15.13%	.539	.558	+3.53%	.104	.269	+158.65%	.708	.682	−3.67%
0.8	.942	.901	+4.35%	.238	.199	+16.39%	.471	.481	+2.12%	.027	.210	+677.78%	.272	.230	−15.44%

Table 18: **Effect of contrastive learning** under MNARL. ‘*w/o*’ refers to using a pre-trained BERT model without fine-tuning. Δ indicates the performance difference between the two settings, and the **bold** highlights the positive improvement. The means across 10 datasets and 10 repeated experiments are reported. $\uparrow$ ($\downarrow$) denotes that higher (lower) is better.

rate	ASMAPE $\downarrow$			KL $\downarrow$			$F_1 \uparrow$			Model $\uparrow$			Feature $\uparrow$		
	<i>w/o</i>	DRIM	Δ	<i>w/o</i>	DRIM	Δ	<i>w/o</i>	DRIM	Δ	<i>w/o</i>	DRIM	Δ	<i>w/o</i>	DRIM	Δ
0.2	.827	.833	−0.73%	.020	.016	+20.00%	.660	.666	+0.91%	.658	.679	+3.19%	.925	.937	+1.30%
0.3	.835	.833	+0.24%	.036	.030	+16.67%	.625	.642	+2.72%	.496	.527	+6.25%	.886	.899	+1.47%
0.4	.839	.849	−1.19%	.054	.050	+7.41%	.591	.611	+3.38%	.326	.442	+35.58%	.832	.841	+1.08%
0.6	.864	.862	+0.23%	.110	.099	+10.00%	.522	.540	+3.45%	.102	.267	+161.76%	.652	.647	−0.77%
0.8	.920	.897	+2.50%	.230	.207	+10.00%	.449	.471	+4.90%	.049	.245	+400.00%	.286	.262	−8.39%

This section examines how the integration of contrastive learning affects the overall performance of DRIM:

- **Fidelity improvements:** When comparing the variant without contrastive learning (denoted as DRIM *w/o* CL) to the DRIM model, there is a consistent reduction in both ASMAPE and KL divergence. This indicates that contrastive learning helps the model fine-tune its representations for better imputation accuracy, especially as the missingness rate increases.
- **Enhanced imputation utility:** In addition to improvements in fidelity, the imputation utility metrics (including F_1 score and model selection performance) are enhanced. Contrastive learning

Table 19: **Effect of contrastive learning** under MNARQ. ‘*w/o*’ refers to using a pre-trained BERT model without fine-tuning. Δ indicates the performance difference between the two settings, and the **bold** highlights the positive improvement. The means across 10 datasets and 10 repeated experiments are reported. $\uparrow$ ($\downarrow$) denotes that higher (lower) is better.

rate	ASMAPE $\downarrow$			KL $\downarrow$			F_1 $\uparrow$			Model $\uparrow$			Feature $\uparrow$		
	<i>w/o</i>	DRIM	Δ	<i>w/o</i>	DRIM	Δ	<i>w/o</i>	DRIM	Δ	<i>w/o</i>	DRIM	Δ	<i>w/o</i>	DRIM	Δ
0.2	.804	.739	+8.08%	.017	.009	+47.06%	.682	.692	+1.47%	.771	.785	+1.82%	.945	.945	0.00%
0.3	.804	.754	+6.22%	.030	.018	+40.00%	.658	.677	+2.89%	.694	.722	+4.03%	.913	.913	0.00%
0.4	.811	.769	+5.18%	.046	.033	+28.26%	.629	.657	+4.45%	.534	.617	+15.54%	.850	.851	+0.12%
0.6	.820	.784	+4.39%	.087	.067	+22.99%	.583	.616	+5.66%	.266	.415	+56.02%	.683	.703	+2.93%
0.8	.844	.781	+7.46%	.161	.124	+22.98%	.515	.560	+8.74%	.151	.290	+92.05%	.545	.536	-1.65%

enables the model to better capture the underlying dependencies between observed and missing data, which in turn leads to improved performance on machine learning tasks.

- Nuanced effects on feature selection: While improvements in feature selection performance are observed, they are sometimes marginal or even slightly negative in certain missingness settings. Nonetheless, the overall effect of contrastive learning is positive and helps improve most metrics significantly.

These results demonstrate that contrastive learning is a key component in the DRIM framework, providing both theoretical justification and empirical evidence of its benefit.

E.5 Sensitivity analysis

We present the additional sensitivity analysis results for missingness rates under MCAR, MAR, MNARL, and MNARQ, respectively.

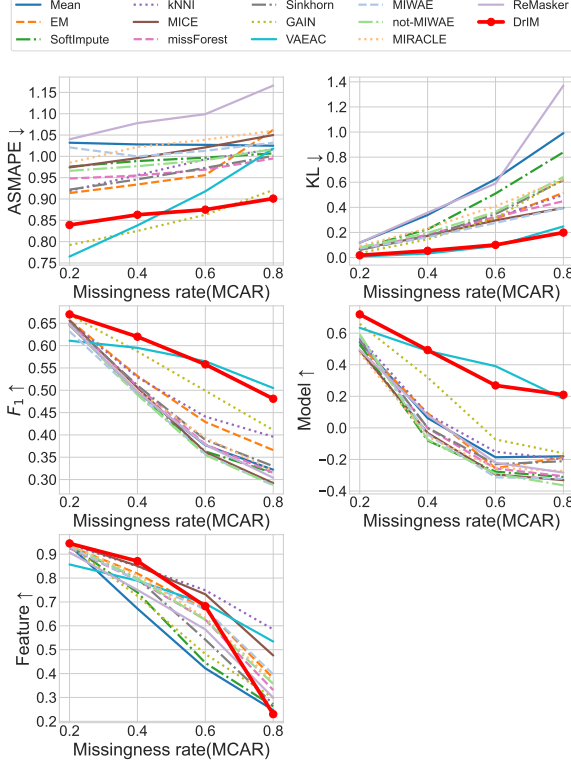


Figure 5: Sensitivity analysis for missingness rates under **MCAR**. Imputation performance comparison of various imputation models under different missing data mechanisms. The means across 10 datasets and 10 repeated experiments are reported. $\uparrow$ ($\downarrow$) denotes higher (lower) is better.

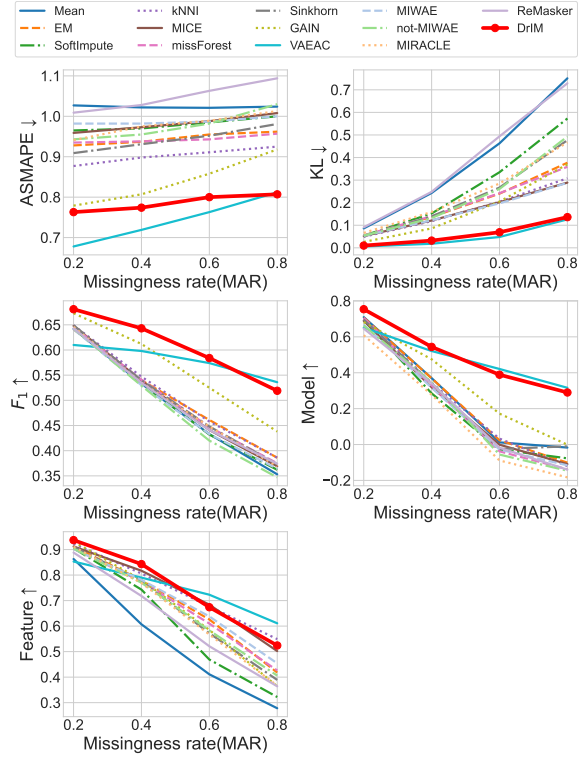


Figure 6: Sensitivity analysis for missingness rates under **MAR**. Imputation performance comparison of various imputation models under different missing data mechanisms. The means across 10 datasets and 10 repeated experiments are reported. $\uparrow$ ($\downarrow$) denotes higher (lower) is better.

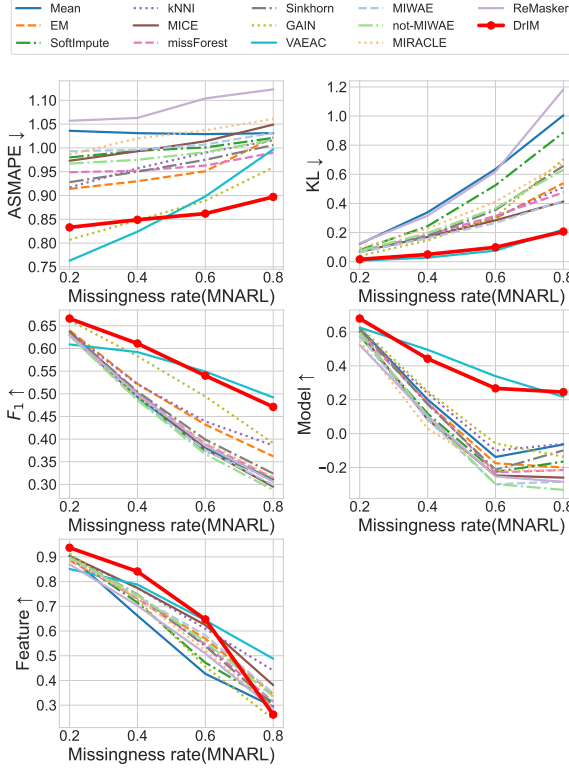


Figure 7: Sensitivity analysis for missingness rates under **MNARL**. Imputation performance comparison of various imputation models under different missing data mechanisms. The means across 10 datasets and 10 repeated experiments are reported. $\uparrow$ ($\downarrow$) denotes higher (lower) is better.

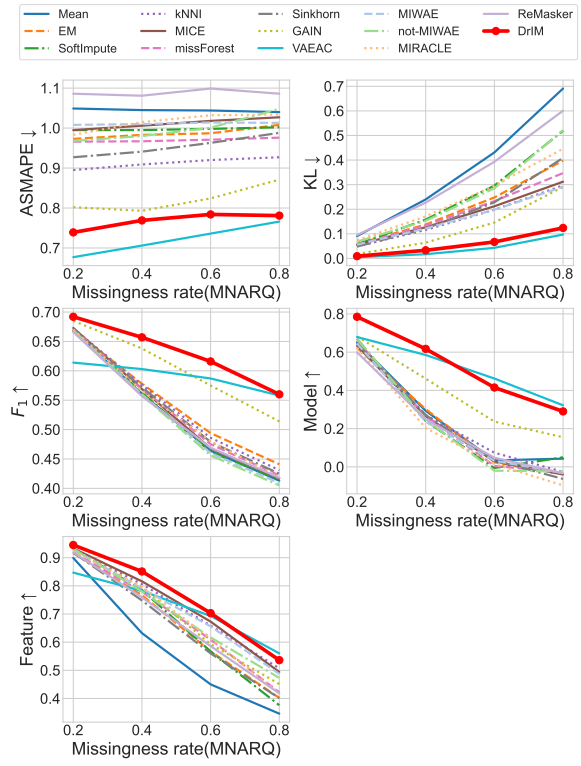


Figure 8: Sensitivity analysis for missingness rates under **MNARQ**. Imputation performance comparison of various imputation models under different missing data mechanisms. The means across 10 datasets and 10 repeated experiments are reported. $\uparrow$ ($\downarrow$) denotes higher (lower) is better.

The final part of Appendix E investigates how sensitive the imputation performance is with respect to varying missingness rates:

- **Impact on imputation fidelity:** As the missingness rate increases, both the ASMAPE and KL divergence generally worsen. However, DRIM consistently maintains lower KL divergence values compared to other methods, indicating its ability to preserve the overall distribution even under harsh conditions.
- **Robustness in imputation utility:** Even as the missingness rate increases, the performance of DRIM in terms of feature selection and classification (F_1 score) remains competitive. Although there is some degradation in performance as missingness becomes severe, DRIM still outperforms many baseline methods.
- **Trade-offs in nearest neighbor estimation:** The sensitivity analysis also highlights the trade-offs associated with the nearest-neighbor-based approach. As more neighbors are included, noise may creep in and negatively impact the imputation. DRIM’s design, enhanced by robust representation and contrastive learning, helps mitigate this issue by carefully balancing the number of neighbors used.

Overall, the sensitivity analysis confirms that DRIM is robust across a range of missingness scenarios, ensuring that it remains effective even when a large proportion of data is missing.

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